On Integration of the Extended Mathematical Programming Formulations for the Travelling Salesman and Vehicle Routing Problems

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Abstract

This study considers two fundamental approaches to formulate the Travelling Salesman (TSP) and many variants of the Capacitated Vehicle Routing (CVRP) problem. Namely, one is based on the label setting approach of Miller et al. (1960) and the other one is based on the one-commodity network flow idea introduced by Gavish and Graves (1978). Contrary to many existing studies comparing and contrasting these approaches, this study introduces a simple idea to integrate them into a combined approach with superior properties in terms of the quality of the linear programming (LP) relaxation. Results of a computational study demonstrate the improved strength of the new formulations numerically compared to those of the underlying formulations, which can be rather significant. One particularly important observation is that while mathematical programming formulations based on the Miller et al. (1960) idea are often weak on their own, the integrated formulations of the same problems are still stronger than the underlying one-commodity network flow formulations. Finally, the computational study provides results of experiments solving various problems using the considered mixed integer linear formulations in a cut-and-branch framework.

Keywords: Travelling Salesman Problem, Vehicle Routing Problem, Time windows constraints, mixed-integer linear programming, Valid inequalities

1. Introduction

Solution algorithms for the Travelling Salesman and Vehicle Routing problems have developed significantly over the last decades. The state of the art solution methodology for the TSP went the long way from solving the problem instance with 49 locations in Dantzig et al. (1954) to solving the instance with 85900 locations in Applegate et al. (2009). The underlying solution methodology for both studies is based on a natural formulation of the problem with an exponential number of constraints that is solved by branch-and-cut. At the same time, the state of the art solution methodology for the VRP developed differently starting from the work of Balinski and Quandt (1964) to the current state of the art in Pessoa et al. (2020). The solution approach is based on the set partitioning representation of the problem with an exponential number of columns, solved by a branch-and-price algorithm. Despite this difference in state of the art methodologies for such related problem, there exist approaches to formulate these problems that simultaneously apply to both problems. This paper specifically focuses on revising such approaches.

One approach initiated by Miller et al. (1960) is based on the introduction of additional variables to the problem formulation that can be interpreted as labels. These variables are not typically used to form the objective function, which is why such an approach leads to a so-called extended formulation of the problem (in the terminology of Pulleyblank (1989)). The labels, instead, are used to form a set of constraints that simultaneously

1. eliminate undesired subtours,

2. allow to impose additional conditions on the route (routes in the VRP case),
3. preserve vehicle capacities (in the VRP case).

A problem formulation that follows the Miller et al. (1960) approach requires only a polynomial number of constraints, which is a major advantage in many situations. Consequently, this approach to formulate the TSP or VRP was improved by Desrochers and Laporte (1991) by lifting the corresponding constraints, making the underlying linear programming (LP) relaxation of the problem stronger.

Another way to approach such problems was proposed by Gavish and Graves (1978) and is based on the idea of a network flow. The flow variables are extra variables introduced to the problem formulation that serve exactly the same purposes listed above. This approach allows us to formulate a problem using a polynomial number of constraints as well.

As two approaches to formulate the problems with similar properties, they have, naturally, attracted attention of researchers to compare them. Wong (1980) demonstrated that the TSP formulation following the Gavish and Graves (1978) approach is stronger than the one following the Miller et al. (1960) approach. When it comes to comparing the Gavish and Graves (1978) TSP formulation with the lifted version of the TSP formulation due to Desrochers and Laporte (1991), Öncan et al. (2009) demonstrated that the two formulations are incomparable. Other extended TSP formulations exist; they can be reviewed, together with further results on their classification and comparison, in Langevin et al. (1990), Gouveia and Voß (1995) and Roberti and Toth (2012).

The motivation of this paper is different and can be stated in terms of the following question: is it possible to integrate these two approaches to model a routing problem instead of comparing and contrasting them? This study answers positively to the question and proposes a simple way to integrate the two incomparable TSP formulations of Desrochers and Laporte (1991) and Gavish and Graves (1978) into one. It turns out that the integrated formulation possesses visibly stronger LP relaxation compared to that of the original formulations. Moreover, the integration idea is valid for the numerous extensions of the TSP that Desrochers and Laporte (1991) and Gavish and Graves (1978) approaches are applicable to. These extensions include the Asymmetric TSP (ATSP) with time windows constraints (ATSP−TW), Asymmetric CVRP (ACVRP), with (ACVRP−TW) or without time windows constraints. Finally, the Heterogeneous CVRP (HCVRP) can be modelled using either of the approaches, therefore the integrated problem formulation of the HCVRP is obtained in this study as well. The paper makes the following contributions:

- the mathematical programming formulations of the ATSP, ATSP−TW, ACVRP, ACVRP−TW and HCVRP based on the label setting and the one-commodity network flow approaches are introduced in a systematic way, along with numerous improvements.
- a new iterative preprocessing scheme for ATSP−TW and ACVRP−TW instances, combining the tightening of arrival time paremeters and elimination of arcs, is introduced.
- the existing in the literature one-commodity network flow formulations of the ATSP−TW and ACVRP−TW (studied in Ascheuer et al. (2001)) are simplified and presented in an equivalent way using roughly n^2 fewer constraints. Likewise, the label setting formulations of ATSP−TW and ACVRP−TW are improved, where the resulting formulations select and combine the stronger inequalities among the default and their lifted versions.
- most central to this study, a general idea to integrate two approaches to formulate the problems is introduced. The idea is valid for all considered problems and results in the integrated formulation for every problem with the improved quality of the LP relaxation.
- the impact of the improved quality of the LP relaxation of the new integrated models is investigated in a computational study.

The paper is organized as follows. Section 2 introduces the ATSP and presents two underlying extended formulations of the problem, followed by the integration idea. Section 3 presents the

ATSP-TW and introduces the adaptations of the two approaches to this problem. Subsequently, Section 4 extends the modelling to the ACVRP case, and Section 5 builds on top of that by handling the time windows constraints in the ACVRP case. Finally, Section 6 considers the HCVRP and develops a new formulation for this problem. Section 7 provides a possible interpretation of the integrated models. Section 8 describes the settings and the results of the computational study; Section 9 concludes.

2. Asymmetric Travelling Salesman Problem

2.1. Extended Mathematical Programming Formulations

Consider a directed graph $G = (V, A)$ with n nodes, $V = \{1, \ldots, n\}$, and the complete set of arcs $A = \{(i, j) \mid i, j \in V, i \neq j\}$. In addition, there exists a matrix $C = \{c_{ij} \geq 0 \mid (i, j) \in A\}$ of distances between the nodes in V. The TSP searches for a Hamiltonian cycle (route) in G where every node is visited once having the smallest possible total distance. With no assumption about the symmetry of matrix C, i.e., that identity $c_{ij} = c_{ji}$ holds for every $(i, j) \in A$, the TSP considered in this section is called the Asymmetric TSP (ATSP).

To model the problem, we introduce a set of binary decision variables $x_{ij} \in \{0, 1\}$ for $(i, j) \in A$ denoting whether arc (i, j) is in a route. The objective function and basic ASTP constraints are presented below:

$$\min \sum_{(i,j)\in A} c_{ij}x_{ij} \tag{2.1}$$

subject to

$$\sum_{j\in V\setminus\{i\}} x_{ij} = \sum_{j\in V\setminus\{i\}} x_{ji} = 1, \qquad i \in V. \tag{2.2}$$

It is known that optimal solutions to the above assignment problem tend to form subtours, which is why proper mathematical programming formulations of the problem include additional constraints to eliminate subtours. The fundamental TSP study of Dantzig et al. (1954) introduced an exponentially sized set of subtour elimination constraints that involved solely x_{ij} variables. Other approaches to eliminate subtours that require extra variables and polynomial number of constraints, exist and constitute the main focus of this study.

The approach of Miller et al. (1960) (referred to as M−T−Z in the following) arbitrary and without loss of generality declares one node as a starting location for a tour (node 1 in our case) and involves additional variables $u_i \geq 0$ (i.e., labels) defined as follows:

$$u_i = k \iff \text{node } i \text{ is } k\text{th in a tour starting from } 1, \qquad i \in V\setminus\{1\}. \tag{2.3}$$

Then, the M−T−Z subtour elimination constraints are as follows:

M−T−Z :

$$u_i - u_j + (n-1)x_{ij} \leq n-2, \qquad i, j \in V\setminus\{1\},\ i \neq j, \tag{2.4}$$

$$u_i \geq 2, \qquad i \in V\setminus\{1\}, \tag{2.5}$$

$$u_i \leq n, \qquad i \in V\setminus\{1\}. \tag{2.6}$$

The original paper of Miller et al. (1960) did not include constraints (2.5)–(2.6) defining bounds on u_i variables as they are, strictly speaking, not necessary. Roberti and Toth (2012) noted that the introduction of such bounds does not affect the LP relaxation quality of the model, but increases the computing time of solving the problem using this formulation. Nevertheless, having these constraints helps to eliminate the ambiguity of defining u_i variables and making the definition (2.3) of variables u_i unique. For this reason, we keep such bounds in all M−T−Z formulations of various problems considered in this paper by default.

Desrochers and Laporte (1991) provided a number of extensions and improvements to the M−T−Z approach for the ATSP. The subtour elimination constraints of Desrochers and Laporte (1991) are called the lifted version of the M−T−Z constraints with additional improvements on the bounds on u_i variables:

D−L :

$$u_i - u_j + (n-1)x_{ij} + (n-3)x_{ji} \leq n-2, \qquad i, j \in V\backslash\{1\},\ i \neq j, \tag{2.7}$$

$$u_i \geq 3 + (n-3)x_{i1} - x_{1i}, \qquad i \in V\backslash\{1\}, \tag{2.8}$$

$$u_i \leq n - 1 - (n-3)x_{1i} + x_{i1}, \qquad i \in V\backslash\{1\}. \tag{2.9}$$

Clearly, D−L constraints are stronger than M−T−Z constraints from the LP relaxation standpoint, while using exactly the same number of extra variables and the same number of constraints. Hence, there are not many reasons to use M−T−Z instead of D−L constraints in practice. The M−T−Z (D−L) constraints is one of the two major approaches to model the ATSP with help of extra variables and a quadratic number of constraints that we consider in this paper.

The idea of Gavish and Graves (1978) for the ATSP is to approach modelling the problem from the network flow perspective. Suppose that the salesman carries $n-1$ units of a commodity upon leaving the depot (node 1), and every location in $V\backslash\{1\}$ demands one unit of that commodity. Therefore, one unit of that commodity is left at every location upon the visit of that location. The commodity is not allowed to flow over other arcs that the salesman does not take. Let the flow variables $\{g_{ij} \geq 0 \mid (i,\, j) \in A,\, i,\, j \neq 1\}$ represent the number of units of the commodity that flows from i to j. The constraints representing this model in a slightly improved form due to Orman and Williams (2006) are presented below:

G−G :

$$(n-1)x_{1i} + \sum_{j \in V\backslash\{1,\, i\}} g_{ji} = \sum_{j \in V\backslash\{1,\, i\}} g_{ij} + 1, \qquad i \in V\backslash\{1\}, \tag{2.10}$$

$$0 \leq g_{ij} \leq (n-2)x_{ij}, \qquad i, j \in V\backslash\{1\},\ i \neq j. \tag{2.11}$$

Strengthening of such constraints is always desirable. Gouveia and Voß (1995) provided a simple observation connecting flow and x_{ij} variables, valid for any integer feasible solution to G−G:

$$g_{ij} \geq x_{ij}, \qquad i, j \in V\backslash\{1\},\ i \neq j. \tag{2.12}$$

The set of valid inequalities (2.12) is quadratic in size and therefore increases the size of the model significantly, if added as is. The computational benefits of having such constraints in the model, in this case, will be problem instance-dependent. Nevertheless, there is a way to incorporate such inequalities implicitly in the model without adding a single extra constraint, proposed in Pavlikov and Petersen (2025). Let the flow from i to j be redefined with help of additional variable $\tilde{g}_{ij}$:

$$g_{ij} = \tilde{g}_{ij} + x_{ij}, \qquad \tilde{g}_{ij} \geq 0. \tag{2.13}$$

This way, flow variables g_{ij} automatically satisfy inequalities (2.12). Representing constraints G−G in terms of $\tilde{g}_{ij}$ variables, the modified version of the G−G constraints appears:

G−G m. :

$$(n-2)x_{1i} + \sum_{j \in V\backslash\{1,\, i\}} \tilde{g}_{ji} = \sum_{j \in V\backslash\{1,\, i\}} \tilde{g}_{ij} + 1 - x_{i1}, \qquad i \in V\backslash\{1\}, \tag{2.14}$$

$$\tilde{g}_{ij} + x_{ij} \leq (n-2)x_{ij} \iff \tilde{g}_{ij} \leq (n-3)x_{ij}, \qquad i, j \in V\backslash\{1\},\ i \neq j, \tag{2.15}$$

$$\tilde{g}_{ij} \geq 0, \qquad i, j \in V\backslash\{1\},\ i \neq j. \tag{2.16}$$

Clearly, the G−G m. formulation is stronger than G−G from the LP relaxation standpoint. Moreover, it requires the same number of constraints and variables as G−G. Therefore, there are not many reasons to use G−G instead of G−G m. in practice.

In order to develop the integrated approach to formulate the problem, note the following observations about the G−G and D−L formulations:

$$\sum_{j \in V \setminus \{1,\, i\}} g_{ij} = n - k \iff \text{node } i \text{ is } k\text{th in a tour starting from } 1, \qquad i \in V \setminus \{1\}, \tag{2.17}$$

$$u_i = k \iff \text{node } i \text{ is } k\text{th in a tour starting from } 1, \qquad i \in V \setminus \{1\}. \tag{2.18}$$

These observations naturally allow for the simple way to integrate G−G and D−L formulations into one with help of the following constraints:

$$\sum_{j \in V \setminus \{1,\, i\}} g_{ij} = n - u_i, \qquad i \in V \setminus \{1\}. \tag{2.19}$$

Likewise, equations connecting the G−G m. and D−L formulations look as follows:

$$\sum_{j \in V \setminus \{1,\, i\}} (\tilde{g}_{ij} + x_{ij}) = \sum_{j \in V \setminus \{1,\, i\}} \tilde{g}_{ij} + 1 - x_{i1} = n - u_i, \qquad i \in V \setminus \{1\}. \tag{2.20}$$

Hence, the integrated ATSP formulation will include extra variables and corresponding constraints simultaneously for G−G m. and D−L models. The models will be connected with help of (2.20) constraints. The full ATSP formulation integrating the one-commodity network flow model (the modified version) and the label-setting model is presented below:

G−G m.+D−L :

$$\min \sum_{(i,\, j) \in A} c_{ij} x_{ij} \tag{2.21}$$

subject to

$$(2.2), (2.7), (2.8), (2.9), \tag{2.22}$$

$$(2.14), (2.15), (2.16), \tag{2.23}$$

$$\sum_{j \in V \setminus \{1,\, i\}} \tilde{g}_{ij} - x_{i1} + u_i = n - 1, \qquad i \in V \setminus \{1\}, \tag{2.24}$$

$$u_i \geq 0, \qquad i \in V \setminus \{1\}, \tag{2.25}$$

$$x_{ij} \in \{0,\, 1\}, \qquad i,\, j \in V,\, i \neq j. \tag{2.26}$$

The **G−G+D−L** formulation can be obtained in a similar way.

2.2. Valid Inequalities

The ATSP has the following important family of valid inequalities that is due to Dantzig et al. (1954):

$$\textbf{D−F−J}: \sum_{i,\, j \in S,\, i \neq j} x_{ij} \leq |S| - 1, \qquad \forall S \subset V,\, 2 \leq |S| \leq n - 1. \tag{2.27}$$

These inequalities can be separated for any values of decision variables $\{x^*_{ij}\}$ (including fractional values) satisfying (2.2) and strengthen any extended ATSP formulation considered in this paper. Separation of such inequalities can be done using algorithms of the polynomial complexity, as described in Crowder and Padberg (1980). A formal description of the separation algorithm and a link to its implementation can be found in Pavlikov and Petersen (2025).

A special subset of the D−F−J inequalities eliminating subtours of size 2 can be notated separately:

$$\textbf{D−F−J(2)}: \; x_{ij} + x_{ji} \leq 1, \qquad\qquad i, j \in V, i < j. \tag{2.28}$$

There exist additional families of valid inequalities for the ATSP, yet their consideration and usage is out of scope of this study.

2.3. Analysis

We start with the following fact that follows immediately from the way the D−L constraints are designed:

$$\text{D−L} \implies \text{D−F−J(2)}. \tag{2.29}$$

In other words, any set of x_{ij} variables satisfying D−L constraints implies the D−F−J(2) inequalities. Let LP$(\cdot)$ denote the optimal value of the LP relaxation of a problem formulation. The following statements follow immediately from the way the new formulations are designed:

$$\text{LP(G−G+D−L)} \geq \text{LP(G−G+D−F−J(2))}, \qquad \text{LP(G−G+D−L)} \geq \text{LP(D−L)}, \tag{2.30}$$

$$\text{LP(G−G m.+D−L)} \geq \text{LP(G−G m.+D−F−J(2))}, \quad \text{LP(G−G m.+D−L)} \geq \text{LP(D−L)}. \tag{2.31}$$

Hence, the new models have superior properties in terms of the quality of the LP relaxation compared to the original formulations. The degree of superiority in numerical terms is investigated in the computational study that follows. However, it is possible to make additional statements comparing the new ATSP formulations with the one proposed in Sherali and Driscoll (2002). The formulation of Sherali and Driscoll (2002) is another extension of the M−T−Z approach obtained via the reformulation-linearization technique. It is less compact than the original M−T−Z formulation, requiring around $4n^2$ constraints. At the same time, it has been demonstrated to be stronger than both G−G and D−L formulations (Öncan et al., 2009). In terms of comparing with the new integrated formulations, the following statements can be made:

$$\text{LP(G−G+D−L)} \leq \text{LP(S−D)}, \tag{2.32}$$

$$\text{LP(G−G m.+D−L)} <> \text{LP(S−D)}. \tag{2.33}$$

Statement (2.32) can be proved examining the formulations directly and noticing that every constraint of G−G+D−L appears in the S−D formulation. Statement (2.33) claiming the incomparability of the two formulations is obtained empirically by observing instances where S−D is stronger and other instances where the G−G m.+D−L formulation is stronger. Result (2.33) is of a special interest since it emphasizes that G−G m.+D−L can be stronger while requiring $2n^2$ less constraints than the S−D formulation.

A more recent study of Pavlikov and Petersen (2025) introduced the two-commodity opposite direction network flow formulations of the ATSP, namely 2CF−ATSP and 2CF−ATSP m. These formulations have been demonstrated to be stronger than the underlying one-commodity network flow counterparts, with the following results proved:

$$\text{LP(2CF−ATSP)} = \text{LP(G−G+D−F−J(2))}, \tag{2.34}$$

$$\text{LP(2CF−ATSP m.)} = \text{LP(G−G m.+D−F−J(2))}. \tag{2.35}$$

In light of (2.30) and (2.31), the following results connect the integrated extended formulations with two-commodity opposite direction network flow formulations of the ATSP:

$$\text{LP(G−G+D−L)} \geq \text{LP(2CF−ATSP)}, \tag{2.36}$$

$$\text{LP(G−G m.+D−L)} \geq \text{LP(2CF−ATSP m.)}. \tag{2.37}$$

3. Asymmetric Travelling Salesman Problem with Time Windows

3.1. Extended Mathematical Programming Formulations

The ATSP with time windows (ATSP−TW) constraints is built on top of the assumptions and notations from the previous section: the complete graph $G = (V, A)$ and the cost matrix C. The new problem is characterized, additionally, by the matrix of travel times $\mathcal{T} = \{\tau_{ij} \geq 0, (i, j) \in A\}$, with the assumption that traversing arc $(i, j) \in A$ takes τ_{ij} units of time. The time windows constraints imply that there exist time intervals $[a_i, b_i] : \ b_i > a_i \geq 0, \ \forall i \in V\backslash\{1\}$, such that the salesman must start the service at node i within time interval $[a_i, b_i]$. On top of that, a set of service time parameters $s_i \geq 0$ denoting the time duration the salesman has to spend at the node $i \in V\backslash\{1\}$ can be present in a problem instance, which we do not explicitly consider in the development of the mathematical programming models. That lack of consideration of service time durations does not limit the generality, though, as service times can be implicitly incorporated into the travel time matrix by redefining travel times as follows $\tau_{ij} := \tau_{ij} + s_i$, $i \in V\backslash\{1\}$. Additionally, a problem instance might have the requirement on the total time of travelling along the tour to be within some maximum allowed value, say T.

The ATSP−TW searches for a least cost Hamiltonian tour over V satisfying the introduced time windows constraints. The problem requires to set the starting position of the salesman as starting location matters for determining the arrival times at other locations; the starting location is node 1 in our notations. Two versions of the ATSP−TW may exist with respect to the possibility of waiting, allowing or not allowing waiting at node if i the salesman arrives earlier than a_i. The latter version is clearly more restrictive. In the following, we consider the version of the problem where waiting is allowed.

The decision variables x_{ij} defining the tour and the objective function are the same as in the case of the ATSP considered in the previous section. Therefore, (2.1)–(2.2) represent the objective function and basic constraints of the ATSP−TW.

The label-setting approach of Miller et al. (1960) is appropriate to model the ATSP−TW as well, once we update the definition of labels u_i. Let $u_i \geq 0$, $i \in V\backslash\{1\}$ be a set of variables defined as follows:

$$u_i = \text{time when the salesman arrives to node } i \text{ in a tour that starts from } 1, \quad i \in V\backslash\{1\}. \tag{3.1}$$

The constraints eliminating subtours and proper tours that do not satisfy time windows constraints are presented below (Ascheuer et al., 2001; Kara and Uzun, 2025):

$$\begin{aligned}
&\mathbf{M-T-Z}: \\
&u_i - u_j + M_{ij}x_{ij} \leq M_{ij} - \tau_{ij}, && i, j \in V\backslash\{1\}, \ i \neq j, && (3.2)\\
&u_i \geq \tau_{1i}x_{1i}, && i \in V\backslash\{1\}, && (3.3)\\
&u_i \geq a_i, && i \in V\backslash\{1\}, && (3.4)\\
&u_i \leq b_i, && i \in V\backslash\{1\}, && (3.5)
\end{aligned}$$

where big M constants are defined as follows:

$$M_{ij} = b_i - a_j + \tau_{ij}, \qquad i, j \in V\backslash\{1\}, \ i \neq j. \tag{3.6}$$

The possible constraint on the maximum tour duration can be incorporated as follows:

$$u_i + \tau_{i1}x_{i1} \leq T, \qquad i \in V\backslash\{1\}. \tag{3.7}$$

Moreover, the remark of Kara and Uzun (2025) emphasizes the importance of having constraints (3.3) in addition to (3.4) in the mathematical programming formulation for the proper solution of the problem.

The lifted version of the above constraints was first introduced in Desrochers and Laporte (1991). Yet, it was later found to be incorrect (Yuan et al., 2020). The following two corrected versions of the lifted M−T−Z constraints for the ATSP−TW were introduced in Yuan et al. (2020):

1. Valid constraints

$$u_i - u_j + Mx_{ij} + (M - \tau_{ij} + a_j - b_i)x_{ji} \leq M - \tau_{ij}, \qquad i, j \in V\backslash\{1\},\ i \neq j, \tag{3.8}$$

2. Supervalid constraints

$$u_i - u_j + Mx_{ij} + \left(M - \tau_{ij} + \mathsf{min}(-\tau_{ji},\, a_j - a_i)\right) x_{ji} \leq M - \tau_{ij}, \quad i, j \in V\backslash\{1\},\ i \neq j, \tag{3.9}$$

where $M = \mathsf{max}_{i,\, j \in V\backslash\{1\}, i \neq j}\, M_{ij}$. Note that if we attempt to convert the first version of the lifted constraints of Yuan et al. (2020) to the model with arc-specific big M values (replacing the single M constant with M_{ij} constants in the corresponding constraints), we obtain the M−T−Z constraints. Given that the advantages of having arc-specific big M values are generally bigger than using a lifted model with larger values of coefficients, we focus on the supervalid version of the constraints with arc-specific big M values only:

$$\begin{aligned}
&\mathbf{Y{-}C{-}O{-}S}: \\
&u_i - u_j + M_{ij}x_{ij} + B_{ji}x_{ji} \leq M_{ij} - \tau_{ij}, && i, j \in V\backslash\{1\},\ i \neq j, && (3.10)\\
&(3.3), (3.4), (3.5), (3.7), && && (3.11)
\end{aligned}$$

where $B_{ji} = b_i - a_j + \mathsf{min}(-\tau_{ji},\, a_j - a_i)$ for $i, j \in V\backslash\{1\},\ i \neq j$.

In the following, we introduce several improvements to the **Y−C−O−S** formulation of the problem. Some of the improvements are new to the literature of time windows constraints.

1. **Improving arrival time parameters**. The time window parameters can be adjusted (narrowed) in a simple way, which affects the efficiency of any problem formulation. Specifically, the earliest and latest arrival times can be redefined as follows

$$\begin{aligned}
&\tilde{a}_i = \mathsf{max}(a_i,\, \tilde{\tau}_{1i}), && (3.12)\\
&\tilde{b}_i = \mathsf{min}(b_i,\, T - \tilde{\tau}_{i1}), \text{where} && (3.13)\\
&\tilde{\tau}_{1i} = \text{ the shortest time to travel from } 1 \text{ to } i \text{ in } G \text{ using } \mathcal{T}, && (3.14)\\
&\tilde{\tau}_{i1} = \text{ the shortest time to travel from } i \text{ to } 1 \text{ in } G \text{ using } \mathcal{T}. && (3.15)
\end{aligned}$$

If travel times in $\mathcal{T}$ satisfy the triangular inequality property, then $\tilde{\tau}_{1i} = \tau_{1i}$ and $\tilde{\tau}_{i1} = \tau_{i1}$. Otherwise, the quantities can be obtained by solving the corresponding shortest path problems.

2. **Improving time window bounds**. Using (3.12) and (3.13), the default constraints (3.4)-(3.5) on the arrival times can be replaced by the stronger inequalities, as suggested by Desrochers and Laporte (1991):

$$\begin{aligned}
&u_i \geq \tilde{a}_i + \sum_{j \in V\backslash\{1,\, i\}} \mathsf{max}(0,\, \tilde{a}_j - \tilde{a}_i + \tau_{ji})x_{ji}, && i \in V\backslash\{1\}, && (3.16)\\
&u_i \leq \tilde{b}_i - \sum_{j \in V\backslash\{1,\, i\}} \mathsf{max}(0,\, \tilde{b}_i - \tilde{b}_j + \tau_{ij})x_{ij}, && i \in V\backslash\{1\}. && (3.17)
\end{aligned}$$

3. **Imroving the initialization constraints (3.3)**. The idea behind (3.16), (3.17) can be used to improve and strengthen (3.3) constraints. First, note that the arrival time to i cannot be not less than $\mathsf{max}(\tilde{a}_i,\, \tau_{1i})$ if $x_{1i} = 1$. If $x_{ji} = 1$, the salesman should first arrive to j (with the earliest arrival time equal to $\tilde{a}_j$) and then proceed to i. Hence, $u_i \geq \tilde{a}_j + \tau_{ji}$ if $x_{ji} = 1$. Note that even if the salesman travels to j right from the depot ($x_{1j} = 1$), eventually it visits i. Hence, the expression $\mathsf{max}(a_j,\, \tau_{1j}) + \tilde{\tau}_{ji}$ provides the lower bound on the time to arrive to i starting from node

j, where

$$\tilde{\tau}_{ji} = \text{ the shortest time to travel from } j \text{ to } i \text{ in } G \text{ using } \mathcal{T}. \tag{3.18}$$

At the same time, $u_i \geq \tilde{a}_i$. The above considerations can be summarized as follows:

$$u_i \geq \begin{cases} \mathsf{max}(a_i,\, \tau_{1i}), \text{ if } x_{1i} = 1, \\ \mathsf{max}(\tilde{a}_i,\, \tilde{a}_j + \tau_{ji}), \qquad \text{if } x_{ji} = 1, \qquad j \in V\backslash\{1,\, i\}, \\ \mathsf{max}(\tilde{a}_i,\, \mathsf{max}(a_j,\, \tau_{1j}) + \tilde{\tau}_{ji}), \qquad \text{if } x_{1j} = 1, \qquad j \in V\backslash\{1,\, i\}. \end{cases} \tag{3.19}$$

Hence, given that $\sum_{j \in V\backslash\{i\}} x_{ji} = 1$ and $\sum_{j \in V\backslash\{1\}} x_{1j} = 1$, the constraints

$$u_i \geq \mathsf{max}(a_i,\, \tau_{1i}) x_{1i} + \sum_{j \in V\backslash\{1,\, i\}} \mathsf{max}(\tilde{a}_i,\, \tilde{a}_j + \tau_{ji}) x_{ji}, \qquad i \in V\backslash\{1\}, \tag{3.20}$$

$$u_i \geq \mathsf{max}(a_i,\, \tau_{1i}) x_{1i} + \sum_{j \in V\backslash\{1,\, i\}} \mathsf{max}(\tilde{a}_i,\, \mathsf{max}(a_j,\, \tau_{1j}) + \tilde{\tau}_{ji}) x_{1j}, \qquad i \in V\backslash\{1\}, \tag{3.21}$$

employ ideas in (5.16) and impose both valid and potentially more accurate lower bounds on the arrival time to i than (3.16).

Likewise, the arrival time to $i \in V\backslash\{1\}$ can be bounded from above as follows:

$$u_i \leq \begin{cases} \mathsf{min}(b_i,\, T - \tau_{i1}), \qquad \text{if } x_{i1} = 1, \\ \mathsf{min}(\tilde{b}_i,\, \tilde{b}_j - \tau_{ij}), \qquad \text{if } x_{ij} = 1, \qquad j \in V\backslash\{1\}, \\ \mathsf{min}(\tilde{b}_i,\, \mathsf{min}(b_j,\, T - \tau_{j1}) - \tilde{\tau}_{ij}), \qquad \text{if } x_{j1} = 1, \qquad j \in V\backslash\{1\}. \end{cases} \tag{3.22}$$

Once again, given that $\sum_{j \in V\backslash\{1,\, i\}} x_{ji} = 1$ and $\sum_{j \in V\backslash\{1\}} x_{j1} = 1$, the constraints

$$u_i \leq \mathsf{min}(b_i,\, T - \tau_{i1}) x_{i1} + \sum_{j \in V\backslash\{1,\, i\}} \mathsf{min}(\tilde{b}_i,\, \tilde{b}_j - \tau_{ij}) x_{ij}, \qquad i \in V\backslash\{1\}, \tag{3.23}$$

$$u_i \leq \mathsf{min}(b_i,\, T - \tau_{i1}) x_{i1} + \sum_{j \in V\backslash\{1,\, i\}} \mathsf{min}(\tilde{b}_i,\, \mathsf{min}(b_j,\, T - \tau_{j1}) - \tilde{\tau}_{ij}) x_{j1}, \qquad i \in V\backslash\{1\}, \tag{3.24}$$

employ ideas in (5.18) and impose both valid and potentially more accurate than (3.17) upper bounds on the arrival time to i.

4. **Improving (3.10) constraints**. With (3.12) and (3.13), the Y−C−O−S constraints can be redefined using the new big M values:

$$\tilde{M}_{ij} = \tilde{b}_i - \tilde{a}_j + \tau_{ij}, \tag{3.25}$$

$$\tilde{B}_{ji} = \tilde{b}_i - \tilde{a}_j + \mathsf{min}(-\tau_{ji},\, \tilde{a}_j - \tilde{a}_i), \qquad i,\, j \in V\backslash\{1\},\, i \neq j. \tag{3.26}$$

Notice that the constant $\tilde{B}_{ji}$ might take a negative value. A negative value of that constant implies that the original M−T−Z constraint is stronger than the new lifted constraint for that particular arc $(i,\, j)$. Therefore, we might construct an improved problem formulation, where the lifted version of the constraint for arc $(i,\, j)$ is used when $B_{ji} > 0$, and the standard M−T−Z constraint is used when $B_{ji} < 0$.

Thus, we obtain the modified Y−C−O−S set of inequalities for preserving time windows constraints:

Y−C−O−S m. :

$$u_i - u_j + \tilde{M}_{ij} x_{ij} + \tilde{B}_{ji} x_{ji} \leq \tilde{M}_{ij} - \tau_{ij}, \qquad i,\, j \in V\backslash\{1\},\, i \neq j : \tilde{B}_{ji} > 0, \tag{3.27}$$

$$u_i - u_j + \tilde{M}_{ij} x_{ij} \leq \tilde{M}_{ij} - \tau_{ij}, \qquad i,\, j \in V\backslash\{1\},\, i \neq j : \tilde{B}_{ji} \leq 0, \tag{3.28}$$

(3.16), (3.17), (3.20), (3.21), (3.23), (3.24), (3.7). (3.29)

Both the M−T−Z and Y−C−O−S m. sets of constraints are known to be weak (from the LP relaxation standpoint) and inefficient for the ATSP−TW. Nevertheless, Y−C−O−S m. is preferable for use if one of the two models must be used.

The one-commodity network flow idea of Gavish and Graves (1978) can be adapted to model the ATSP−TW with a slight modification that is due to van Eijl (1995). Consider the flow of the "time" commodity, where flow variable y_{ij} is defined as follows:

$$y_{ij} = \begin{cases} \text{time to arrive to } i, & \text{if } x_{ij} = 1, \\ 0, & \text{otherwise,} \end{cases} \quad i \in V\backslash\{1\},\, j \in V,\, i \neq j. \tag{3.30}$$

Then, the one-commodity network flow constraints for the ATSP−TW using the updated $\tilde{a}_i$ and $\tilde{b}_i$ defined according to (3.12) and (3.13) are as follows:

G−G−E :

$$\mathsf{max}(a_i,\, \tau_{1i}x_{1i}) + \sum_{j\in V\backslash\{1,\,i\}} y_{ji} + \sum_{j\in V\backslash\{1,\,i\}} \tau_{ji}x_{ji} \leq \sum_{j\in V\backslash\{i\}} y_{ij}, \qquad i \in V\backslash\{1\}, \tag{3.31}$$

$$y_{ij} \geq \tilde{a}_i x_{ij}, \qquad i \in V\backslash\{1\},\, j \in V,\, i \neq j, \tag{3.32}$$

$$y_{ij} \leq \tilde{b}_i x_{ij}, \qquad i \in V\backslash\{1\},\, j \in V,\, i \neq j, \tag{3.33}$$

$$y_{ij} \geq 0, \qquad i \in V\backslash\{1\},\, j \in V,\, i \neq j. \tag{3.34}$$

The maximum tour duration requirement in this time flow framework can be implemented with help of the new flow variables in the following way:

$$\sum_{i\in V\backslash\{1\}} (y_{i1} + \tau_{i1}x_{i1}) \leq T. \tag{3.35}$$

Using the same idea behind the G−G m. formulation of the ATSP, we can improve the G−G−E constraints by eliminating a substantial number of constraints without deteriorating their LP relaxation quality. For that, we redefine the y_{ij} variables with help of $\tilde{y}_{ij} \geq 0$ in the following way:

$$y_{ij} = \tilde{y}_{ij} + \tilde{a}_i x_{ij}, \qquad i \in V\backslash\{1\},\, j \in V,\, i \neq j, \tag{3.36}$$

Thus, y_{ij} variables automatically satisfy the (3.32) constraints. The improved version of the G−G−E constraints is obtained:

$$\mathsf{max}(a_i,\, \tau_{1i}x_{1i}) + \sum_{j\in V\backslash\{1,\,i\}} (\tilde{y}_{ji} + \tilde{a}_j x_{ji}) + \sum_{j\in V\backslash\{1,\,i\}} \tau_{ji}x_{ji} \leq$$

$$\sum_{j\in V\backslash\{i\}} (\tilde{y}_{ij} + \tilde{a}_i x_{ij}) = \sum_{j\in V\backslash\{i\}} \tilde{y}_{ij} + \tilde{a}_i, \qquad i \in V\backslash\{1\}, \tag{3.37}$$

$$\tilde{y}_{ij} \leq (\tilde{b}_i - \tilde{a}_i)x_{ij}, \qquad i \in V\backslash\{1\},\, j \in V,\, i \neq j, \tag{3.38}$$

$$\tilde{y}_{ij} \geq 0, \qquad i \in V\backslash\{1\},\, j \in V,\, i \neq j. \tag{3.39}$$

The following additional improvements to the formulation can be developed. Consider the arrival time to i represented by the expression $\tilde{y}_{ij} + \tilde{a}_i$ in a tour where $j \in V\backslash\{1\}$ is the next location after i. The latest arrival time to i is $\tilde{b}_i$, however, it cannot exceed $\tilde{b}_j - \tau_{ij}$ due to the restriction of travelling to j after i. Therefore, the updated upper bound on the arrival time to i can be written down as follows:

$$\tilde{y}_{ij} + \tilde{a}_i \leq \mathsf{min}(\tilde{b}_i,\, \tilde{b}_j - \tau_{ij}),$$

if $\tilde{a}_i \leq \mathsf{min}(\tilde{b}_i,\, \tilde{b}_j - \tau_{ij})$. Otherwise, if $\tilde{a}_i > \mathsf{min}(\tilde{b}_i,\, \tilde{b}_j - \tau_{ij})$, one can conclude that the salesman can not travel from i and j and satisfy time windows constraints at the same time, which implies $x_{ij} = 0$. If the salesman travels to depot after i, i.e., if $x_{i1} = 1$, the arrival time to i is bounded by the minimum value of the deadline $\tilde{b}_i$ and the maximum possible time $T - \tau_{i1}$, sufficient to finish the tour within T units of time. These observations allow us to create the following improved version of G−G−E:

G−G−E m. :

$$\mathsf{max}(a_i,\, \tau_{1i}x_{1i}) + \sum_{j \in V\setminus\{1,\, i\}} (\tilde{y}_{ji} + \tilde{a}_j x_{ji} + \tau_{ji}x_{ji}) \leq$$

$$\tilde{a}_i + \sum_{j \in V\setminus\{i\}} \tilde{y}_{ij}, \qquad i \in V\setminus\{1\}, \tag{3.40}$$

$$\tilde{y}_{ij} \leq \Big(\mathsf{min}(\tilde{b}_i,\, \tilde{b}_j - \tau_{ij}) - \tilde{a}_i\Big)\, x_{ij}, \qquad i,\, j \in V\setminus\{1\},\, i \neq j, \tag{3.41}$$

$$\tilde{y}_{i1} \leq \Big(\mathsf{min}(\tilde{b}_i,\, T - \tau_{i1}) - \tilde{a}_i\Big)\, x_{i1}, \qquad i \in V\setminus\{1\}, \tag{3.42}$$

$$\tilde{y}_{ij} \geq 0, \qquad i \in V\setminus\{1\},\, j \in V,\, i \neq j. \tag{3.43}$$

The G−G−E m. constraints are preferred to use in comparison to G−G−E due to improved efficiency and strength.

The integrated formulation that combines the one-commodity time flow model G−G−E m. and a label setting approach is based upon the following observation for $i \in V\setminus\{1\}$:

$$\sum_{j \in V\setminus\{1\}} y_{ij} = \tilde{a}_i + \sum_{j \in V\setminus\{1\}} \tilde{y}_{ij} = \text{time when the salesman arrives to } i = u_i. \tag{3.44}$$

Therefore, the integrated formulation of basic models G−G−E and Y−C−O−S is obtained:

G−G−E+Y−C−O−S :

$$\min_{x_{ij},\, \tilde{y}_{ij},\, u_i} \sum_{(i,j) \in A} c_{ij}x_{ij} \tag{3.45}$$

subject to

(2.2), (3.31), (3.32), (3.33), (3.34), (3.35),

(3.10), (3.11),

$$u_i = \sum_{j \in V\setminus\{i\}} y_{ij}, \qquad i \in V\setminus\{1\}, \tag{3.46}$$

$$x_{ij} \in \{0,\, 1\}, \qquad i,\, j \in V,\, i \neq j. \tag{3.47}$$

We integrate next the G−G−E m. and Y−C−O−S m. constraints as being the most efficient ones within their own classes of constraints and obtain the following new ATSP−TW formulation:

G−G−E m.+Y−C−O−S m. :

$$\min_{x_{ij},\, \tilde{y}_{ij},\, u_i} \sum_{(i,j) \in A} c_{ij}x_{ij} \tag{3.48}$$

subject to

(2.2), (3.40), (3.41), (3.42), (3.43),

(3.27), (3.28), (3.29),

$$u_i = a_i + \sum_{j \in V\setminus\{i\}} \tilde{y}_{ij}, \qquad i \in V\setminus\{1\}, \tag{3.49}$$

$$x_{ij} \in \{0, 1\}, \qquad i, j \in V, i \neq j. \tag{3.50}$$

3.2. Preprocessing and Valid Inequalities

Preprocessing is an extremely important step in designing efficient ATSP−TW formulations. Just as careful definition of big M constants employed while designing problem formulations in the previous subsection, tightening of time windows themselves is equally important. The following time windows preprocessing routines are available (Ascheuer et al., 2001):

- Earliest arrival time adjustments:

$$\tilde{a}_i := \mathsf{max}\left(\tilde{a}_i, \min_{j \in V\setminus\{1,i\}} (\tilde{a}_j + \tau_{ji})\right), \qquad i \in V\setminus\{1\}, \tag{3.51}$$

$$\tilde{a}_i := \mathsf{max}\left(\tilde{a}_i, \mathsf{min}\left(\tilde{b}_i, \min_{j \in V\setminus\{1,i\}} (\tilde{a}_j - \tau_{ij})\right)\right), \qquad i \in V\setminus\{1\}. \tag{3.52}$$

- Latest arrival time adjustments:

$$\tilde{b}_i := \mathsf{min}\left(\tilde{b}_i, \mathsf{max}\left(\tilde{a}_i, \max_{j \in V\setminus\{1,i\}} \left(\tilde{b}_j + \tau_{ji}\right)\right)\right), \qquad i \in V\setminus\{1\}, \tag{3.53}$$

$$\tilde{b}_i := \mathsf{min}\left(\tilde{b}_i, \max_{j \in V\setminus\{1,i\}} \left(\tilde{b}_j - \tau_{ij}\right)\right), \qquad i \in V\setminus\{1\}. \tag{3.54}$$

In addition, the following straightforwardly motivated arc and corresponding variable elimination can be done:

$$x_{ij} = 0, \text{ if } \tilde{a}_i + \tau_{ij} > \tilde{b}_j \text{ or if } \mathsf{min}(\tilde{b}_i, \tilde{b}_j - \tau_{ij}) < \tilde{a}_i, \tag{3.55}$$

$$x_{i1} = 0, \text{ if } \tilde{a}_i + \tau_{i1} > T, \tag{3.56}$$

$$x_{1i} = 0, \text{ if } \tau_{1i} > \tilde{b}_i. \tag{3.57}$$

However, the variable fixing and time window adjustments can be and often are interconnected. In other words, an adjustment of the earliest or latest arrival time may trigger the elimination of a variable, while the elimination of a variable may correspondingly lead to an additional adjustment of arrival time deadlines. Therefore, what we introduce and implement is the iterative preprocessing scheme, where arrival deadline adjustments and variable elimination is done until neither of them can be done. The formal algorithm is outlined in Algorithm 1.

In terms of valid inequalities, we note that the same family of D−F−J inequalities (2.27) is valid to the ATSP−TW and applicable to strengthen any ATSP−TW formulation, even though such inequalities are not specific to any particular set of time windows intervals. Additional results on valid inequalities specifically designed for the ATSP−TW can be found in Ascheuer et al. (2001).

3.3. Analysis

As for the ATSP case, the new integrated models have superior properties in terms of the quality of the LP relaxation compared to that of the original formulations:

$$\text{LP(G−G−E m.+Y−C−O−S m.)} \geq \text{LP(G−G−E m.)}, \tag{3.58}$$

$$\text{LP(G−G−E m.+Y−C−O−S m.)} \geq \text{LP(Y−C−O−S m.)}. \tag{3.59}$$

The degree of superiority in numerical terms is investigated in the computational experiments section.

Algorithm 1 ATSP−TW Preprocessing

$o := 1$ # denotes the depot
$A_e := \{(i, o) \,|\, i \in V\backslash\{o\},\ \tilde{a}_i + \tau_{io} > T\} \cup \{(o, i) \,|\, i \in V\backslash\{o\},\ \tau_{oi} > \tilde{b}_i\}$
Input: (V, A, A_e), o, $\mathcal{T}$, T, $\{\tilde{a}_i, \tilde{b}_i, \tilde{\tau}_{oi}, \tilde{\tau}_{io} : i \in V\backslash\{o\}\}$
continue := $True$
while continue **do**:
    count := 0
    for $i \in V\backslash\{o\}$ **do**
        $a_i := \max\left(\min_{j:(j,i)\in A\backslash A_e,\, j\neq o}(\tilde{a}_j + \tau_{ji}),\ \min\left(\tilde{b}_i,\ \min_{j:(i,j)\in A\backslash A_e,\, j\neq o}(\tilde{a}_j - \tau_{ij})\right)\right)$
        if $a_i > \tilde{a}_i$ **then**
            $\tilde{a}_i := a_i$
            count $+= 1$
        end if
        $b_i := \min\left(\max\left(\tilde{a}_i,\ \max_{j:(j,i)\in A\backslash A_e, j\neq o}\left(\tilde{b}_j + \tau_{ji}\right)\right),\ \max_{j:(i,j)\in A\backslash A_e, j\neq o}\left(\tilde{b}_j - \tau_{ij}\right)\right)$
        if $b_i < \tilde{b}_i$ **then**
            $\tilde{b}_i := b_i$
            count $+= 1$
        end if
        for $j \in V\backslash\{o, i\}$ **do**
            if $\tilde{a}_i + \tau_{ij} > \tilde{b}_j$ OR $\min(\tilde{b}_i, \tilde{b}_j - \tau_{ij}) < \tilde{a}_i$ **then**
                $A_e := A_e \cup \{(i, j)\}$
                count $+= 1$
            end if
        end for
    end for
    if count $= 0$ **then**
        continue := $False$
    end if
end while
$A_e := A_e \cup \{(i, o) \in A\backslash A_e,\ \tilde{a}_i + \tau_{io} > T\} \cup \{(o, i) \in A\backslash A_e,\ \tau_{oi} > \tilde{b}_i\}$
return A_e, $\{\tilde{a}_i, \tilde{b}_i, i \in V\backslash\{o\}\}$

4. Asymmetric Capacitated Vehicle Routing Problem

4.1. Extended Mathematical Programming Formulations

The Capacitated Vehicle Routing problem (CVRP) deals with a set of customers $\mathcal{C}$ that require the delivery (or pick up) of a certain commodity. Each customer $i \in \mathcal{C}$ requires $d_i > 0$ units of that commodity to be delivered (or picked up); we assume in this study that delivery (pick up) of the demanded number of units of the commodity cannot be split among several vehicles. To organize the delivery, a number of vehicles located at a depot location denoted by 0 is available; vehicles must return to the depot once customers are served. Vehicles are assumed to have the capacity of $Q > 0$ to carry that commodity. Therefore, delivery routes must satisfy the vehicle capacity constraint. The objective is to minimize the total cost of the delivery routes used by the fleet of vehicles. To evaluate the cost of every routing decision, an asymmetric matrix of costs (distances) between locations $C = \{c_{ij} \mid i, j \in \mathcal{C}_0,\ i \neq j\}$ is available, with $\mathcal{C}_0 = \mathcal{C} \cup \{0\}$. Due to the asymmetry assumption on the matrix C, the problem is referred to as the Asymmetric CVRP (ACVRP).

To model the ACVRP, we first introduce the directed network $G = (\mathcal{C}_0, A)$ where A is the complete set of arcs in G, $A = \{(i, j) \,|\, i, j \in \mathcal{C}_0,\ i \neq j\}$. Subsequently, the set of decision variables $x_{ij} \in \{0, 1\}$ for $(i, j) \in A$ is created, where x_{ij} defines whether a vehicle traverses arc (i, j). The

objective function and basic ACVRP constraints requiring that every customer in $\mathcal{C}$ must be visited by a vehicle once are presented below:

$$\min \sum_{(i,j)\in A} c_{ij}x_{ij} \tag{4.1}$$

subject to

$$\sum_{j\in\mathcal{C}_0\setminus\{i\}} x_{ij} = \sum_{j\in\mathcal{C}_0\setminus\{i\}} x_{ji} = 1, \qquad i \in \mathcal{C}. \tag{4.2}$$

As before, we present the label-based approach of Miller et al. (1960) to formulate the problem first. In this approach, we assume that demand d_i represents the number of units of the commodity to be picked up at customer i. Let the variable $u_i \geq 0$ denote the cumulative load of a vehicle that visits customer i upon leaving i. Then, constraints that simultaneously eliminate subtours not connected to the depot and preserve vehicle capacities are presented below:

M−T−Z :

$$u_i - u_j + Qx_{ij} \leq Q - d_j, \qquad i, j \in \mathcal{C}, i \neq j, \tag{4.3}$$

$$u_i \geq d_i, \qquad i \in \mathcal{C}, \tag{4.4}$$

$$u_i \leq Q, \qquad i \in \mathcal{C}. \tag{4.5}$$

The lifted version of the above constraints for the ACVRP was presented in Desrochers and Laporte (1991), subject to a correction in Kara et al. (2004):

D−L :

$$u_i - u_j + Qx_{ij} + (Q - d_i - d_j)x_{ji} \leq Q - d_j, \qquad i, j \in \mathcal{C}, i \neq j, \tag{4.6}$$

$$u_i \geq d_i + \sum_{j\in\mathcal{C}\setminus\{i\}} d_j x_{ji}, \qquad i \in \mathcal{C}, \tag{4.7}$$

$$u_i \leq Q - (Q - \max_{j\in\mathcal{C}\setminus\{i\}} d_j - d_i)x_{0i} - \sum_{j\in\mathcal{C}\setminus\{i\}} d_j x_{ij}, \qquad i \in \mathcal{C}. \tag{4.8}$$

As usual for the label-setting approach, the D−L version of constraints for the ACVRP is more advanced than M−T−Z both from theoretical and practical standpoints.

The one-commodity network flow approach of Gavish and Graves (1978) for the ACVRP is created as follows. Suppose that every vehicle carries Q units of commodity upon leaving the depot. Upon visiting customer i, d_i units of that commodity are released from the vehicle, i.e., every customer i gets the delivery of d_i units of the commodity. To model this idea, a set of flow variables f_{ij} = remaining number of units of the commodity in a vehicle when that vehicle leaves i and travels to j via arc (i, j) is introduced. Then, the flow balance constraints preserving vehicle capacities and eliminating subtours not connected to the depot are as follows:

G−G :

$$Qx_{0i} + \sum_{j\in\mathcal{C}} f_{ji} = \sum_{j\in\mathcal{C}_0} f_{ij} + d_i, \qquad i \in \mathcal{C}, \tag{4.9}$$

$$f_{ij} \leq Qx_{ij}, \qquad i, j \in \mathcal{C}, i \neq j, \tag{4.10}$$

$$f_{ij} \geq 0, \qquad i \in \mathcal{C}, j \in \mathcal{C}_0, i \neq j. \tag{4.11}$$

Gouveia (1995) introduced the following set of stronger inequalities instead of (4.10):

$$f_{ij} \geq d_j x_{ij}, \quad f_{ij} \leq (Q - d_i)x_{ij}, \qquad i, j \in \mathcal{C}, i \neq j. \tag{4.12}$$

Employing the familiar transformation of variables $f_{ij} = \tilde{f}_{ij} + d_j x_{ij}$, $\tilde{f}_{ij} \geq 0$, the modified and strengthened version of the **G−G** constraints appears:

G−G m. :

$$Qx_{0i} + \sum_{j \in \mathcal{C} \setminus \{i\}} \left(\tilde{f}_{ji} + d_i x_{ji} \right) = \sum_{j \in \mathcal{C} \setminus \{i\}} \left(\tilde{f}_{ij} + d_j x_{ij} \right) + f_{i0} + d_i, \qquad i \in \mathcal{C}, \tag{4.13}$$

$$\tilde{f}_{ij} \leq (Q - d_i - d_j) x_{ij}, \qquad i, j \in \mathcal{C}, i \neq j, \tag{4.14}$$

$$\tilde{f}_{ij} \geq 0, \qquad i, j \in \mathcal{C}, i \neq j, \tag{4.15}$$

$$f_{i0} \geq 0, \qquad i \in \mathcal{C}. \tag{4.16}$$

Thus, the G−G m. constraints are superior to G−G in terms of the LP relaxation strength while requiring the same number of variables and equations / inequalities.

To integrate the two approaches for the ACVRP, note that the total outflow of the commodity at customer i, $\sum_{j \in \mathcal{C} \setminus \{i\}} \left(\tilde{f}_{ij} + d_j x_{ij} \right) + f_{i0}$, equals the remaining number of units of the commodity on the vehicle when that vehicle leaves i. The remaining number of units of the commodity in the vehicle, on the other hand, should be equal to the initial number of units of the commodity when that vehicle left the depot less the cumulative number of delivered units of that commodity in the label-setting modelling framework. In other words, the total outflow of the commodity at location i in the G−G m. model equals $Q - u_i$ in the notations of the Miller et al. (1960) framework for the ACVRP. Hence, the integrated ACVRP mathematical programming model appears:

G−G m.+D−L :

$$\min_{x_{ij},\, \tilde{f}_{ij},\, f_{i0},\, u_i} \sum_{(i,j) \in A} c_{ij} x_{ij} \tag{4.17}$$

subject to

$(4.2), (4.13), (4.14), (4.15), (4.16),$

$(4.6), (4.7), (4.8),$

$$u_i = Q - \sum_{j \in \mathcal{C} \setminus \{i\}} \left(\tilde{f}_{ij} + d_j x_{ij} \right) - f_{i0},\ u_i \geq 0, \qquad i \in \mathcal{C}, \tag{4.18}$$

$$x_{ij} \in \{0, 1\}, \qquad i, j \in \mathcal{C}_0, i \neq j. \tag{4.19}$$

In a similar way the G−G+D−L formulation of the ACVRP can be obtained.

4.2. Valid Inequalities

Valid inequalities appropriate to the ATSP preventing subtours that are not connected to the depot and are therefore valid to the ACVRP as well. Subtours of size 3 and larger are eliminated with help of the following inequalities:

$$\mathbf{D-F-J}: \ x\left(S, \bar{S}\right) = \sum_{i \in S} \sum_{j \in \mathcal{C}_0 \setminus \{S\}} x_{ij} \geq 1, \qquad \forall S \subset \mathcal{C}, |S| \geq 3. \tag{4.20}$$

Subtours of size 2 are eliminated using the simpler set of inequalities:

$$\mathbf{D-F-J(2)}: \ x_{ij} + x_{ji} \leq 1, \qquad i, j \in \mathcal{C}, i < j. \tag{4.21}$$

The above inequalities do not take into account the capacity of the vehicles. The ACVRP allows for many families of valid inequalities employing the vehicle capacity as a parameter. We only consider a subset of families presented in Letchford and Salazar-González (2006) that we can use or those we can associate directly with the developments in this paper.

One family of the vehicle capacity-based inequalities is the set of Fractional Capacity (FC) inequalities:

$$\textbf{FCI}: \; x\left(S, \bar{S}\right) \geq \sum_{i \in S} d_i/Q, \qquad \forall S \subset \mathcal{C}, \, |S| \geq 2, \tag{4.22}$$

that deal with capacities of the corresponding cut-sets $x\left(S, \bar{S}\right)$ and impose the minimum possible number of vehicles that leaves (enters) S. That minimum number of vehicles, clearly, is determined based on the capacity of the vehicles Q. Such inequalities can be separated in a polynomial time. However, it is also known (Gouveia, 1995) that the simple one-commodity network flow ACVRP formulation we consider and use in this paper implies this set of inequalities, which is why we do not deal with separation of such inequalities.

A more accurate family of inequalities for the ACVRP, called the Rounded Capacity (RC) inequalities, was introduced by Laporte and Nobert (1983), and looks as follows:

$$\textbf{RCI}: \; x\left(S, \bar{S}\right) \geq \left\lceil \sum_{i \in S} d_i/Q \right\rceil, \qquad \forall S \subset \mathcal{C}, \, |S| \geq 2. \tag{4.23}$$

The RC inequalities are a direct improvement of the corresponding FC inequalities where the right-hand side of the inequality is rounded up. Due to this rounding up operation, the separation problem of these inequalities becomes NP-hard (Diarrassouba, 2017). Therefore, various heuristic algorithms for separation of these inequalities exist (Ralphs et al., 2003; Lysgaard et al., 2004). At the same time, these inequalities can be often separated exactly and efficiently using mixed integer linear programming techniques and a corresponding solver, see Pavlikov et al. (2024).

Finally, another relevant to this study family of valid inequalities for the CVRP exist, called the Generalized Multistar inequalities (Gouveia, 1995):

$$\textbf{GMI}: \; x\left(S, \bar{S}\right) \geq \frac{1}{Q} \sum_{i \in S} d_i + \sum_{i \in S} \sum_{j \in \mathcal{C}_0 \setminus \{S\}} d_j (x_{ij} + x_{ji}), \qquad \forall S \subset \mathcal{C}, \, |S| \geq 2. \tag{4.24}$$

4.3. Analysis

First, we mention the known results due to Gouveia (1995):

$$\text{G}-\text{G} \implies \text{FCI}, \tag{4.25}$$

$$\text{G}-\text{G m.} \implies \text{GMI}, \tag{4.26}$$

and the following result available immediately by inspection of the D−L constraints:

$$\text{D}-\text{L} \implies \text{D}-\text{F}-\text{J}(2). \tag{4.27}$$

The integrated ACVRP model is stronger than G−G m. and D−L models in terms of the optimal values of their LP relaxations and the following more accurate results can be stated:

$$\text{LP}(\text{G}-\text{G m.}+\text{D}-\text{L}) \geq \text{LP}(\text{G}-\text{G m.}+\text{D}-\text{F}-\text{J}(2)), \tag{4.28}$$

$$\text{LP}(\text{G}-\text{G m.}+\text{D}-\text{L}) \geq \text{LP}(\text{D}-\text{L}+\text{GMI}). \tag{4.29}$$

5. Asymmetric Capacitated Vehicle Routing Problem with Time Windows

5.1. Extended Mathematical Programming Formulations

The ACVRP with time windows constraints (ACVRP−TW) is an important practical extension of the ACVRP. In addition to the cost matrix C and vehicle capacity Q to carry the demanded commodity, the problem assumes a matrix of travel times $\mathcal{T} = \{\tau_{ij} \mid (i, j) \in A\}$ between all pairs of

nodes, including the depot. Serving a customer is assumed to take some service time $s_i \geq 0$, $i \in \mathcal{C}$, therefore a vehicle leaves the customer s_i time units later than the arrival time. With service times explicitly present in the problem description, the effective travel time matrix is convenient to introduce $\mathcal{T}' = \{s_i + \tau_{ij} \mid (i, j) \in A\}$, assuming that $s_0 = 0$.

The ACVRP−TW implies that every customer $i \in \mathcal{C}$ has a time window interval $[a_i, b_i]$ when a vehicle should arrive to serve this customer. One important consequence of such additional constraints on the arrival times is that even if the cost matrix C and the corresponding time matrix $\mathcal{T}'$ are symmetric, these problems are typically modeled as directed network problems (Toth and Vigo, 2002). Additionally, the problem might have a requirement that every route can not take longer than a prespecified time duration, say T.

To model the ACVRP−TW, we introduce a set of decision variables $x_{ij} \in \{0, 1\}$ for $(i, j) \in A$, where x_{ij} defines whether a vehicle traverses arc (i, j). The objective function and basic ACVRP−TW constraints requiring that every customer in $\mathcal{C}$ must be visited by a vehicle once are the same as in the ACVRP case (4.1)-(4.2). In addition, a problem formulation should have the vehicle capacity preserving constraints, which can be G−G m., D−L or G−G m.+D−L constraints introduced in the previous section. Thus, the following paragraphs discuss how the time windows constraints can be developed in the ACVRP context.

The label-setting approach of Miller et al. (1960) is appropriate to model the time windows constraints for the ACVRP−TW case in a similar way as it was developed for the ATSP−TW. Let $u_i \geq 0$, $i \in \mathcal{C}$ be a set of variables defined as follows:

$$u_i = \text{time when a vehicle arrives to node } i \text{ in a route that starts from node } 0, \quad i \in \mathcal{C}. \tag{5.1}$$

The constraints eliminating subtours and proper tours that do not satisfy time windows constraints are presented below (Ascheuer et al., 2001; Kara and Uzun, 2025):

M−T−Z :

$$u_i - u_j + M_{ij}x_{ij} \leq M_{ij} - \tau_{ij}, \qquad i, j \in \mathcal{C}, i \neq j, \tag{5.2}$$

$$u_i \geq \tau_{0i}x_{0i}, \qquad i \in \mathcal{C}, \tag{5.3}$$

$$u_i \geq a_i, \qquad i \in \mathcal{C}, \tag{5.4}$$

$$u_i \leq b_i, \qquad i \in \mathcal{C}, \tag{5.5}$$

where big M constants are defined as follows:

$$M_{ij} = b_i - a_j + \tau_{ij} + s_i, \qquad i, j \in \mathcal{C}, i \neq j. \tag{5.6}$$

As pointed out in the ATSP−TW section, constraints (5.3) have not typically been part of a problem formulation in existing studies, while Kara and Uzun (2025) emphasise their importance for a correct solution of the problem. The possible additional requirement on the maximum duration for every route can be implemented as follows:

$$u_i + s_i + \tau_{i0}x_{i0} \leq T, \qquad i \in \mathcal{C}. \tag{5.7}$$

The lifted version of the constraints is immediately available from (Yuan et al., 2020; Kara and Uzun, 2025) as well:

Y−C−O−S :

$$u_i - u_j + M_{ij}x_{ij} + B_{ji}x_{ji} \leq M_{ij} - \tau_{ij}, \qquad i, j \in \mathcal{C}, i \neq j, \tag{5.8}$$

$$(5.3), (5.4), (5.5), (5.7), \tag{5.9}$$

where $B_{ji} = b_i - a_j + \min(-\tau_{ji} - s_j, a_j - a_i)$ for $i, j \in \mathcal{C}$, $i \neq j$.

Following the development of the improved Y−C−O−S constraints for the ATSP−TW, we

introduce the similar modification of the constraints for the ACVRP−TW.

1. **Improving arrival time parameters**. The time window parameters can be adjusted (narrowed) in a simple way, which affects the efficiency of any problem formulation. Specifically, the earliest and latest arrival times can be redefined as follows

$$\tilde{a}_i = \mathsf{max}(a_i,\, \tilde{\tau}_{0i}), \tag{5.10}$$

$$\tilde{b}_i = \mathsf{min}(b_i,\, T - \tilde{\tau}_{i0}), \text{where} \tag{5.11}$$

$$\tilde{\tau}_{0i} = \text{ the shortest time to travel from } 0 \text{ to } i \text{ in } G \text{ using } \mathcal{T}', \tag{5.12}$$

$$\tilde{\tau}_{i0} = \text{ the shortest time to travel from } i \text{ to } 0 \text{ in } G \text{ using } \mathcal{T}'. \tag{5.13}$$

If travel times in $\mathcal{T}'$ satisfy the triangular inequality property, then $\tilde{\tau}_{0i} = \tau_{0i}$ and $\tilde{\tau}_{i0} = \tau_{i0}$. Otherwise, the quantities can be obtained by solving the corresponding shortest path problems.

2. **Improving time window bounds**. Using (5.10) and (5.11), the default constraints (5.4)-(5.5) on the arrival times can be replaced by the stronger inequalities, as suggested by Desrochers and Laporte (1991):

$$u_i \geq \tilde{a}_i + \sum_{j \in \mathcal{C}\setminus\{i\}} \mathsf{max}(0,\, \tilde{a}_j - \tilde{a}_i + s_j + \tau_{ji}) x_{ji}, \qquad i \in \mathcal{C}, \tag{5.14}$$

$$u_i \leq \tilde{b}_i - \sum_{j \in \mathcal{C}\setminus\{i\}} \mathsf{max}(0,\, \tilde{b}_i - \tilde{b}_j + s_i + \tau_{ij}) x_{ij}, \qquad i \in \mathcal{C}. \tag{5.15}$$

3. **Imroving the initialization constraints (5.3)**. The idea behind (5.14), (5.15) can be used to improve and strengthen (5.3) constraints. First, note that the arrival time to i cannot be not less than $\mathsf{max}(a_i,\, \tau_{0i})$ if $x_{0i} = 1$. If $x_{ji} = 1$, the salesman should first arrive to j (with the earliest arrival time equal to $\tilde{a}_j$) and then proceed to i. Hence, $u_i \geq \tilde{a}_j + s_j + \tau_{ji}$ if $x_{ji} = 1$. Still, $u_i \geq \tilde{a}_i$. The above considerations are summarized as follows:

$$u_i \geq \begin{cases} \mathsf{max}(a_i,\, \tau_{0i}), \text{ if } x_{0i} = 1, \\ \mathsf{max}(\tilde{a}_i,\, \tilde{a}_j + s_j + \tau_{ji}), \qquad \text{if } x_{ji} = 1, \qquad j \in \mathcal{C}\setminus\{i\}. \end{cases} \tag{5.16}$$

Hence, given that $\sum_{j \in \mathcal{C}_0\setminus\{i\}} x_{ji} = 1$, the constraints

$$u_i \geq \mathsf{max}(a_i,\, \tau_{0i}) x_{0i} + \sum_{j \in \mathcal{C}\setminus\{i\}} \mathsf{max}(\tilde{a}_i,\, \tilde{a}_j + s_j + \tau_{ji}) x_{ji}, \qquad i \in \mathcal{C}, \tag{5.17}$$

employ ideas in (5.16) and impose both valid and potentially more accurate lower bounds on the arrival time to i than (5.14).

Likewise, the arrival time to $i \in \mathcal{C}$ can be bounded from above as follows:

$$u_i \leq \begin{cases} \mathsf{min}(b_i,\, T - \tau_{i0}), \qquad \text{if } x_{i0} = 1, \\ \mathsf{min}(\tilde{b}_i,\, \tilde{b}_j - s_i - \tau_{ij}), \qquad \text{if } x_{ij} = 1, \qquad j \in \mathcal{C}. \end{cases} \tag{5.18}$$

Once again, given that $\sum_{j \in \mathcal{C}_0\setminus\{i\}} x_{ji} = 1$, the constraints

$$u_i \leq \mathsf{min}(b_i,\, T - \tau_{i0}) x_{i0} + \sum_{j \in \mathcal{C}\setminus\{i\}} \mathsf{min}(\tilde{b}_i,\, \tilde{b}_j - s_i - \tau_{ij}) x_{ij}, \qquad i \in \mathcal{C}, \tag{5.19}$$

employ the ideas of (5.18) and impose both valid and potentially more accurate than (5.15) upper bounds on the arrival time to i.

4. **Improving (5.8) constraints**. With (5.10) and (5.11), the Y−C−O−S constraints can be

redefined using the new big M values:

$$\tilde{M}_{ij} = \tilde{b}_i - \tilde{a}_j + s_i + \tau_{ij}, \tag{5.20}$$

$$\tilde{B}_{ji} = \tilde{b}_i - \tilde{a}_j + \mathsf{min}(-s_j - \tau_{ji},\, \tilde{a}_j - \tilde{a}_i), \qquad i,\, j \in \mathcal{C},\, i \neq j. \tag{5.21}$$

Notice that the constant $\tilde{B}_{ji}$ might take a negative value. A negative value of that constant implies that the original M−T−Z constraint is stronger than the new lifted constraint for that particular arc $(i,\, j)$. Therefore, we can construct an improved problem formulation, where the lifted version of the constraint for arc $(i,\, j)$ is used when $B_{ji} > 0$, and the standard M−T−Z constraint is used when $B_{ji} < 0$:

Y−C−O−S m. :

$$u_i - u_j + \tilde{M}_{ij} x_{ij} + \tilde{B}_{ji} x_{ji} \leq \tilde{M}_{ij} - \tau_{ij}, \qquad i,\, j \in \mathcal{C},\, i \neq j : \tilde{B}_{ji} > 0, \tag{5.22}$$

$$u_i - u_j + \tilde{M}_{ij} x_{ij} \leq \tilde{M}_{ij} - \tau_{ij}, \qquad i,\, j \in \mathcal{C},\, i \neq j : B_{ji} \leq 0, \tag{5.23}$$

$$(5.14), (5.15)(5.17), (5.19). \tag{5.24}$$

The introduced M−T−Z, Y−C−O−S or and Y−C−O−S m. models are known to be weak (from the LP relaxation standpoint) and inefficient for the ACVRP−TW. Nevertheless, Y−C−O−S m. is preferable for use if one of the formulations must be used.

The one-commodity network flow idea of Gavish and Graves (1978) can be adapted to model the ACVRP−TW with a slight modification that is due to van Eijl (1995). Consider once again the flow of the "time" commodity, where flow variable y_{ij} is defined as follows:

$$y_{ij} = \begin{cases} \text{time to arrive to } i, & \text{if } x_{ij} = 1, \\ 0, & \text{otherwise,} \end{cases} \qquad i \in \mathcal{C},\, j \in \mathcal{C}_0,\, i \neq j. \tag{5.25}$$

Then, the one-commodity network flow constraints for the ACVRP−TW are as follows:

G−G−E :

$$\mathsf{max}(\tau_{0i},\, a_i) x_{0i} + \sum_{j \in \mathcal{C} \setminus \{i\}} y_{ji} + \sum_{j \in \mathcal{C} \setminus \{i\}} (s_j + \tau_{ji}) x_{ji} \leq \sum_{j \in \mathcal{C}_0 \setminus \{i\}} y_{ij}, \qquad i \in \mathcal{C}, \tag{5.26}$$

$$y_{ij} \geq \tilde{a}_i x_{ij}, \qquad i \in \mathcal{C},\, j \in \mathcal{C}_0,\, i \neq j, \tag{5.27}$$

$$y_{ij} \leq \tilde{b}_i x_{ij}, \qquad i \in \mathcal{C},\, j \in \mathcal{C}_0,\, i \neq j, \tag{5.28}$$

$$y_{ij} \geq 0, \qquad i \in \mathcal{C},\, j \in \mathcal{C}_0,\, i \neq j. \tag{5.29}$$

The possible maximum route duration requirement can be implemented with help of the time flow variables as follows:

$$y_{i0} + s_i + \tau_{i0} x_{i0} \leq T, \qquad i \in \mathcal{C}. \tag{5.30}$$

Using the same idea behind the G−G m. formulation for the ATSP, we can improve the G−G−E constraints by eliminating a substantial number of constraints without deteriorating their LP relaxation quality. For that, we redefine the y_{ij} variables with help of $\tilde{y}_{ij} \geq 0$ in the following way:

$$y_{ij} = \tilde{y}_{ij} + \tilde{a}_i x_{ij}, \qquad i \in \mathcal{C},\, j \in \mathcal{C}_0,\, i \neq j. \tag{5.31}$$

Thus, y_{ij} variables automatically satisfy the (5.27) constraints. The first improved version of the

G−G−E constraints is now obtained:

$$\mathsf{max}(\tau_{0i},\, a_i)x_{0i} + \sum_{j\in\mathcal{C}\setminus\{i\}} (\tilde{y}_{ji} + \tilde{a}_j x_{ji}) + \sum_{j\in\mathcal{C}\setminus\{i\}} (s_j + \tau_{ji})x_{ji} \leq$$
$$\sum_{j\in\mathcal{C}_0\setminus\{i\}} (\tilde{y}_{ij} + \tilde{a}_i x_{ij}) = \tilde{a}_i + \sum_{j\in\mathcal{C}_0\setminus\{i\}} \tilde{y}_{ij}, \qquad i\in\mathcal{C}, \qquad (5.32)$$

$$\tilde{y}_{ij} \leq (\tilde{b}_i - \tilde{a}_i)x_{ij}, \qquad i\in\mathcal{C},\, j\in\mathcal{C}_0,\, i\neq j, \qquad (5.33)$$

$$\tilde{y}_{ij} \geq 0, \qquad i\in\mathcal{C},\, j\in\mathcal{C}_0,\, i\neq j. \qquad (5.34)$$

The following additional improvement to the constraints above can be developed. Consider the arrival time $\tilde{y}_{ij} + \tilde{a}_i$ to i in a route where $j\in\mathcal{C}$ is the next location after i. The latest arrival time to i is $\tilde{b}_i$, however, it cannot exceed $\tilde{b}_j - s_i - \tau_{ij}$ due to the restriction of travelling to j after i. Therefore, the updated upper bound on the arrival time to i can be written down as follows:

$$\tilde{y}_{ij} + \tilde{a}_i \leq \mathsf{min}(\tilde{b}_i,\, \tilde{b}_j - s_i - \tau_{ij}).$$

When a vehicle continues to depot following i, i.e., when $x_{i0} = 1$, the latest arrival time to i might be affected by the maximum allowed route duration, as follows:

$$\tilde{y}_{i0} + \tilde{a}_i \leq \mathsf{min}(\tilde{b}_i,\, T - \tau_{i0} - s_i).$$

These observation allows us to create the following improved version of the G−G−E constraints:

G−G−E m. :

$$\mathsf{max}(\tau_{0i},\, a_i)x_{0i} + \sum_{j\in\mathcal{C}\setminus\{i\}} (\tilde{y}_{ji} + (\tilde{a}_j + s_j + \tau_{ji})x_{ji}) \leq \tilde{a}_i + \sum_{j\in\mathcal{C}_0\setminus\{i\}} \tilde{y}_{ij}, \qquad i\in\mathcal{C}, \quad (5.35)$$

$$\tilde{y}_{ij} \leq \Big(\mathsf{min}(\tilde{b}_i,\, \tilde{b}_j - \tau_{ij} - s_i) - \tilde{a}_i\Big)\, x_{ij}, \qquad i\in\mathcal{C},\, j\in\mathcal{C},\, i\neq j, \quad (5.36)$$

$$\tilde{y}_{i0} \leq \Big(\mathsf{min}(\tilde{b}_i,\, T - \tau_{i0} - s_i) - \tilde{a}_i\Big)\, x_{i0}, \qquad i\in\mathcal{C}, \quad (5.37)$$

$$\tilde{y}_{ij} \geq 0, \qquad i\in\mathcal{C},\, j\in\mathcal{C}_0,\, i\neq j. \quad (5.38)$$

The integrated formulation that combines the one-commodity time flow models and the label setting approache for the ACVRP−TW is based upon the following observations for $i\in\mathcal{C}$:

$$\sum_{j\in\mathcal{C}_0\setminus\{i\}} y_{ij} = \tilde{a}_i + \sum_{j\in\mathcal{C}_0\setminus\{i\}} \tilde{y}_{ij} = \text{time when the salesman arrives to } i = u_i. \qquad (5.39)$$

Hence, the integration of the G−G−E and Y−C−O−S formulations is done with help of the following constraints:

$$u_i = \sum_{j\in\mathcal{C}_0\setminus\{i\}} y_{ij}, \qquad i\in\mathcal{C}, \qquad (5.40)$$

while the integration of the improved versions, G−G−E m. and Y−C−O−S m., is done with help of the following constraints:

$$u_i = \tilde{a}_i + \sum_{j\in\mathcal{C}_0\setminus\{i\}} \tilde{y}_{ij}, \qquad i\in\mathcal{C}. \qquad (5.41)$$

We introduce two integrated formulations next. One is obtained by integrating the basic formulations G−G−E and Y−C−O−S:

G−G−E+Y−C−O−S :

$$\min \sum_{(i,j)\in A} c_{ij}x_{ij} \tag{5.42}$$

subject to

$$(4.2), \text{G}-\text{G m., D}-\text{L or G}-\text{G m.}+\text{D}-\text{L ACVRP capacity constraints}, \tag{5.43}$$

$$(5.8), (5.9), (5.26), (5.27), (5.28), (5.29), \tag{5.44}$$

$$u_i = \sum_{j\in\mathcal{C}_0\setminus\{i\}} y_{ij},\ u_i \geq 0, \qquad i \in \mathcal{C}, \tag{5.45}$$

$$x_{ij} \in \{0,\ 1\}, \qquad i,\ j \in \mathcal{C}_0,\ i \neq j. \tag{5.46}$$

The more efficient integrated formulation is obtained by integrating the improved formulations G−G−E m. and Y−C−O−S m. as being the most efficient formulations within their own classes, we obtain the strongest integrated ACVRP−TW formulation:

G−G−E m.+Y−C−O−S m. :

$$\min \sum_{(i,j)\in A} c_{ij}x_{ij} \tag{5.47}$$

subject to

$$(4.2), \text{G}-\text{G m., D}-\text{L or G}-\text{G m.}+\text{D}-\text{L ACVRP capacity constraints}, \tag{5.48}$$

$$(5.22), (5.23), (5.24), (5.35), (5.36), (5.37), (5.38), \tag{5.49}$$

$$u_i = \tilde{a}_i + \sum_{j\in\mathcal{C}_0\setminus\{i\}} \tilde{y}_{ij},\ u_i \geq 0, \qquad i \in \mathcal{C}, \tag{5.50}$$

$$x_{ij} \in \{0,\ 1\}, \qquad i,\ j \in \mathcal{C}_0,\ i \neq j. \tag{5.51}$$

5.2. Preprocessing and Valid Inequalities

As in the ATSP−TW case considered before, preprocessing is an extremely important step in designing efficient ACVRP−TW formulations. Preprocessing routines similar to those described (Ascheuer et al., 2001) are presented here, adapted to the notations of the ACVRP−TW:

- Earliest arrival time adjustments:

$$\tilde{a}_i := \max\left(\tilde{a}_i,\ \min_{j\in\mathcal{C}\setminus\{i\}}\left(\tilde{a}_j + s_j + \tau_{ji}\right)\right), \qquad i \in \mathcal{C}, \tag{5.52}$$

$$\tilde{a}_i := \max\left(\tilde{a}_i,\ \min\left(\tilde{b}_i,\ \min_{j\in\mathcal{C}\setminus\{i\}}\left(\tilde{a}_j - s_i - \tau_{ij}\right)\right)\right), \qquad i \in \mathcal{C}. \tag{5.53}$$

- Latest arrival time adjustments:

$$\tilde{b}_i := \min\left(\tilde{b}_i,\ \max\left(\tilde{a}_i,\ \max_{j\in\mathcal{C}\setminus\{i\}}\left(\tilde{b}_j + s_j + \tau_{ji}\right)\right)\right), \qquad i \in \mathcal{C}, \tag{5.54}$$

$$\tilde{b}_i := \min\left(\tilde{b}_i,\ \max_{j\in\mathcal{C}\setminus\{i\}}\left(\tilde{b}_j - s_i - \tau_{ij}\right)\right), \qquad i \in \mathcal{C}. \tag{5.55}$$

In addition, the following straightforwardly motivated variable elimination can be done for $(i,\ j) \in A,\ i \neq 0,\ j \neq 0$:

$$x_{ij} = 0, \text{ if } \tilde{a}_i + s_i + \tau_{ij} > \tilde{b}_j \text{ or if } \min(\tilde{b}_i,\ \tilde{b}_j - s_i - \tau_{ij}) < \tilde{a}_i, \tag{5.56}$$

$$x_{ij} = 0, \text{ if } d_i + d_j > Q. \tag{5.57}$$

An arc connecting $i \in \mathcal{C}$ can be eliminated upon the following condition

$$x_{i0} = 0, \text{ if } \tilde{a}_i + \tilde{\tau}_{i0} > T, \qquad i \in \mathcal{C}. \tag{5.58}$$

As in the ATSP− TW case, tightening of arrival time parameters and elimination of arcs can impact each other and, therefore, the iterative preprocessing algorithm developed for the ATSP−TW is applicable for the ACVRP−TW almost directly. Subject to the change of notations and a slightly different way of defining the initial set of eliminated arcs A_e for the ACVRP−TW, the preprocessing algorithm for the ACVRP−TW simply calls the Algorithm 1, detailed as follows:

Algorithm 2 ACVRP−TW Preprocessing

1: $o := 0$
2: Input: $(\mathcal{C}_0, A)$, $\mathcal{T}'$, T, $\{\tilde{a}_i, \tilde{b}_i, \tilde{\tau}_{oi}, \tilde{\tau}_{io}, i \in \mathcal{C}\}$, Q
3: $\mathcal{C} := \mathcal{C} \backslash \{o\}$
4: $A_e := \{(i, o) \,|\, i \in \mathcal{C}, \tilde{a}_i + \tilde{\tau}_{io} > T\} \cup \{(i, j) \in A \,|\, d_i + d_j > Q, i \neq o, j \neq o\}$
5: Execute lines #4 – 30 of Algorithm 1

In terms of applicable valid inequalities, we note that all families of valid inequalities developed for the ACVRP are valid and applicable to strengthen any ACVRP−TW formulation, even though such inequalities are not specific to any particular set of time windows intervals. Additional results on valid inequalities that can be adapted to handle the time windows constraints of the ACVRP−TW can be found in Ascheuer et al. (2001).

5.3. Analysis

As for the ATSP−TW case, the new integrated models have superior properties in terms of the quality of the LP relaxation compared to that of the original formulations:

$$\text{LP(G−G−E m.+Y−C−O−S m.)} \geq \text{LP(G−G−E m.)}, \tag{5.59}$$

$$\text{LP(G−G−E m.+Y−C−O−S m.)} \geq \text{LP(Y−C−O−S m.)}. \tag{5.60}$$

The degree of superiority in numerical terms is investigated in the computational experiments section.

6. Heterogeneous Capacitated Vehicle Routing Problem

6.1. Extended Mathematical Programming Formulations

The Heterogeneous Capacitated Vehicle Routing Problem (HCVRP) is a natural extension of the standard ACVRP. As in the ACVRP case, the problem is characterized by a set $\mathcal{C}$ of customers with $d_i > 0$ units of a commodity for pick up and a depot location 0 where vehicles are located to deliver this commodity. As before, we adopt the notation $\mathcal{C}_0 = \mathcal{C} \cup \{0\}$. However, the vehicles are no longer assumed homogeneous, but rather have the heterogeneous capacities to carry the commodity. Specifically, we assume that there is a set $\mathcal{K}$, $|\mathcal{K}| = K$, of vehicle types, where a vehicle of type $k \in \mathcal{K}$ has the capacity Q_k to carry the commodity. Without loss of generality, we assume that $Q_1 > Q_2 > \ldots > Q_K > 0$. Moreover, we assume that there is a limited fleet available to organize this particular delivery, with parameters $\mu_k \in \mathbb{Z}_+$ denoting how many vehicles of type k are available to deliver the commodity to customers; $\bar{\mu} = (\mu_1, \ldots, \mu_K)$. When a vehicle of type k is employed in the delivery, payment of the fixed charge fee of $F_k \geq 0$ is involved. Therefore, the objective of the problem is to simultaneously select which vehicle types to engage into service, how many vehicles of each type to engage and which routes these vehicles should take to minimize the total cost. The total cost, hence, consists of the transportation cost component and the fixed cost component.

To model the HCVRP, we introduce a set of decision variables $x_{ij} \in \{0, 1\}$ for $i, j \in \mathcal{C}_0$, $i \neq j$ denoting whether a vehicle of any type traverses arc (i, j). In addition, we need to introduce some extra variables:

$$a_{ik} = \begin{cases} 1, & \text{if } i \in \mathcal{C} \text{ is the last location in a route used by a vehicle of type } k, \\ 0, & \text{otherwise.} \end{cases} \tag{6.1}$$

Thus, variables a_{ik} can be used to extract the vehicle-engagement decisions. Indeed, the expression $\sum_{i \in \mathcal{C}} a_{ik}$ represents the number of vehicles of type k involved in the delivery.

The objective function and basic HCVRP constraints requiring that every customer in $\mathcal{C}$ must be visited by a vehicle once and involved vehicles are within capacities are then presented as follows:

$$\min \sum_{k \in \mathcal{K}} F_k \sum_{i \in \mathcal{C}} a_{ik} + \sum_{(i,j) \in A} c_{ij} x_{ij} \tag{6.2}$$

subject to

$$x_{i0} = \sum_{k \in \mathcal{K}} a_{ik}, \qquad i \in \mathcal{C}, \tag{6.3}$$

$$\sum_{i \in \mathcal{C}} a_{ik} \leq \mu_k, \qquad k \in \mathcal{K}, \tag{6.4}$$

$$\sum_{j \in \mathcal{C}_0 \setminus \{i\}} x_{ij} = \sum_{j \in \mathcal{C}_0 \setminus \{i\}} x_{ji} = 1, \qquad i \in \mathcal{C}. \tag{6.5}$$

As usual, we start with a mathematical programming model for the HCVRP based on the Miller et al. (1960) approach. Such a model was developed in Yaman (2006). Let label $u_i > 0$ denote the cumulative load of a vehicle that visits customer i and picks up the customer's requests for the commodity, upon leaving i. Then, constraints that simultaneously eliminate subtours not connected to the depot and preserve vehicle capacities are presented below:

$\mathbf{Y}$:

$$u_i - u_j + Q_1 x_{ij} + \left(\min_{k \in \mathcal{K}_{ij}} Q_k - q_i - q_j \right) x_{ji} \leq Q_1 - d_j - \sum_{k \in \mathcal{K}_i} (Q_1 - Q_k) a_{ik}, \qquad i, j \in \mathcal{C},\ i \neq j, \tag{6.6}$$

$$u_i \geq d_i + \sum_{j \in \mathcal{C} \setminus \{i\}} d_j x_{ji}, \qquad i \in \mathcal{C}, \tag{6.7}$$

$$u_i \leq \sum_{k \in \mathcal{K}_i} Q_k a_{ik} + Q_1 \left(1 - \sum_{k \in \mathcal{K}_i} a_{ik} \right) - \sum_{j \in \mathcal{C} \setminus \{i\}} d_j x_{ij} - \min \left(Q_1 - d_i - \max_{j \in \mathcal{C} \setminus \{i\}} d_j,\ \min_{k \in \mathcal{K}_i} Q_k - d_i \right) x_{0i}, \qquad i \in \mathcal{C}, \tag{6.8}$$

where

$$\mathcal{K}_{ij} = \{k \in \mathcal{K} : d_i + d_j \leq Q_k\}, \qquad i, j \in \mathcal{C},\ i \neq j, \tag{6.9}$$

$$\mathcal{K}_i = \{k \in \mathcal{K} : d_i \leq Q_k\}, \qquad i \in \mathcal{C}. \tag{6.10}$$

To introduce the one-commodity network flow model for the HCVRP, we need a set of flow variables $f_{ij} \geq 0$, for $(i, j) \in A$. As we mentioned earlier, a model can exist in two equivalent versions, in the pick up or delivery version. Typically, the delivery version of the formulation is used to introduce the one-commodity network flow model for the ACVRP. However, in order to

integrate the model further with the label-setting formulation in the HCVRP settings, we have to use the pick up version of the model. The model is created as follows: we imagine that customers have $d_i > 0$ units of a commodity that needs to be picked up by vehicles. Vehicles leave the depot empty and the flow variable f_{ij} denote the number of units of the commodity in a vehicle when it leaves i and travels to j via (i, j) arc. Then, the one-commodity network flow constraints for the HCVRP are as follows:

$$\mathbf{G{-}G}:$$

$$d_i + \sum_{j \in \mathcal{C} \setminus \{i\}} f_{ji} = \sum_{j \in \mathcal{C}_0 \setminus \{i\}} f_{ij}, \qquad i \in \mathcal{C}, \tag{6.11}$$

$$f_{ij} \leq Q_1 x_{ij}, \qquad i, j \in \mathcal{C}, i \neq j, \tag{6.12}$$

$$f_{i0} \leq \sum_{k \in \mathcal{K}} Q_k a_{ik}, \qquad i \in \mathcal{C}. \tag{6.13}$$

Valid inequalities of Gouveia (1995) apply to the HCVRP as well, subject to the adjustment due to the pick up orientation of the flow:

$$d_i x_{ij} \leq f_{ij} \leq (Q_1 - d_j) x_{ij}, \qquad (i, j) \in A. \tag{6.14}$$

Using the flow variable transformation employed previously:

$$f_{ij} = \tilde{f}_{ij} + d_i x_{ij}, \qquad \tilde{f}_{ij} \geq 0, \tag{6.15}$$

we obtain the modified one-commodity network flow problem formulation with pick up orientation, equivalent to the delivery-based model used in Pavlikov (2025):

$$\mathbf{G{-}G\ m.}:$$

$$d_i + \sum_{j \in \mathcal{C} \setminus \{i\}} \left(\tilde{f}_{ji} + d_j x_{ji} \right) = \sum_{j \in \mathcal{C} \setminus \{i\}} \left(\tilde{f}_{ij} + d_i x_{ij} \right) + f_{i0}, \qquad i \in \mathcal{C}, \tag{6.16}$$

$$\tilde{f}_{ij} \leq (Q_1 - d_j - d_i) x_{ij}, \qquad i, j \in \mathcal{C}, i \neq j, \tag{6.17}$$

$$f_{i0} \leq \sum_{k \in \mathcal{K}} Q_k a_{ik}, \qquad i \in \mathcal{C}. \tag{6.18}$$

With this, the integration of $\mathbf{G{-}G\ m.}$ and $\mathbf{Y}$ formulations is based upon the following set of equations:

$$u_i = \sum_{j \in \mathcal{C} \setminus \{i\}} \left(\tilde{f}_{ij} + d_i x_{ij} \right) + f_{i0}, \qquad i \in \mathcal{C}. \tag{6.19}$$

The integrated HCVRP formulation appears:

$$\mathbf{G{-}G\ m.{+}Y}:$$

$$\min \sum_{k \in \mathcal{K}} F_k \sum_{i \in \mathcal{C}} a_{ik} + \sum_{(i,j) \in A} c_{ij} x_{ij} \tag{6.20}$$

subject to

(6.3), (6.4), (6.5),

(6.16), (6.17), (6.18),

(6.6), (6.7), (6.8),

$$u_i = \sum_{j \in \mathcal{C} \setminus \{i\}} \left(\tilde{f}_{ij} + d_i x_{ij} \right) + f_{i0}, \qquad i \in \mathcal{C}, \tag{6.21}$$

$$a_{ik} \in \{0, 1\}, \qquad i \in \mathcal{C},\ k \in \mathcal{K}, \tag{6.22}$$
$$x_{i0} \geq 0, \qquad i \in \mathcal{C}, \tag{6.23}$$
$$x_{ij} \in \{0, 1\}, \qquad i \in \mathcal{C}_0,\ j \in \mathcal{C},\ i \neq j. \tag{6.24}$$

6.2. Valid Inequalities

We can use several families of valid inequalities for the HCVRP to strengthen every considered problem formulation. Baldacci et al. (2009) introduced the following adaptation of the Rounded Capacity inequalities for the HCVRP:

B−B−V RCI :

$$x\left(S, \bar{S}\right) + \sum_{h=1}^{k} \left\lceil \frac{Q_h}{Q_{k+1}} - 1 \right\rceil \sum_{i \in \mathcal{C}} a_{ih} \geq \left\lceil \frac{\sum_{i \in S} d_i}{Q_{k+1}} \right\rceil, \quad k = 1, \ldots, K-1,\ \ S \subseteq \mathcal{C},\ |S| \geq 2, \tag{6.25}$$

where the minimum capacity of the cut-set $\left(S, \bar{S}\right)$ depends on the fleet engagement decisions $\sum_{i \in \mathcal{C}} a_{ih}$.

Pavlikov (2025) proposed two additional families of valid inequalities for the mathematical programming formulations of the problem. One family is yet another version of the Rounded Capacity inequalities, where the minimum capacity of the cut-set $\left(S, \bar{S}\right)$ is independent of vehicle engagement decisions:

$$\textbf{RCI}: \ x\left(S, \bar{S}\right) \geq m_{\vec{Q}, \vec{\mu}}(S), \qquad S \subseteq \mathcal{C},\ |S| \geq 2, \tag{6.26}$$

where $m_{\vec{Q}, \vec{\mu}}(S)$ provides a lower bound on the minimum number of vehicles required to serve customers in S. The bound can be obtained as follows

$$m_{\vec{Q}, \vec{\mu}}(S) = \min_{z_k} \sum_{k \in \mathcal{K}} z_k$$
subject to
$$\sum_{k \in \mathcal{K}} Q_k z_k \geq \sum_{i \in S} d_i,$$
$$z_k \leq \mu_k, \qquad k \in \mathcal{K}, \tag{6.27}$$
$$z_k \in \mathbb{Z}_+, \qquad k \in \mathcal{K}. \tag{6.28}$$

Another family of inequalities introduces constraints on the minimum capacities of the cut-sets that take fractional values with help of additional auxiliary variables:

$$\textbf{FCI}: \ x\left(S, \bar{S}\right) \geq \min_{z_k^S} \sum_{k \in \mathcal{K}} z_k^S, \qquad S \subseteq \mathcal{C},\ |S| \geq 2, \tag{6.29}$$
$$\sum_{k \in \mathcal{K}} Q_k z_k^S \geq \sum_{i \in S} d_i, \qquad S \subseteq \mathcal{C},\ |S| \geq 2, \tag{6.30}$$
$$z_k^S \leq \sum_{i \in \mathcal{C}} a_{ik}, \qquad S \subseteq \mathcal{C},\ |S| \geq 2,\ k \in \mathcal{K}, \tag{6.31}$$
$$z_k^S \geq 0, \qquad S \subseteq \mathcal{C},\ |S| \geq 2,\ k \in \mathcal{K}. \tag{6.32}$$

These inequalities by design are dependent on fleet engagement decisions $\sum_{i \in \mathcal{C}} a_{ik}$. All violated inequalities from all three families can be identified exactly and efficiently using exact separation routines developed in Pavlikov (2025).

7. Further Interpretation Details

This section offers a possible interpretation of the nature of the integrated extended mathematical programming formulations. The proposed interpretation will be illustrated on the basis of the ATSP, but it extends to formulations of other problems as well.

Consider the key set of constraints implementing the integrated formulation G−G m.+D−L of the ATSP:

$$\sum_{j \in V \setminus \{1,\, i\}} \tilde{g}_{ij} - x_{i1} + u_i = n - 1, \qquad i \in V \setminus \{1\}. \tag{7.1}$$

The constraints suggest that u_i variables can be represented via corresponding expressions involving flow variables $\tilde{g}_{ij}$ and routing variables x_{ij}. This way, the integrated ATSP formulation, formally speaking, does not require variables u_i and can be constructed using flow variables only. This observation suggests that D−L inequalities may serve as a yet another set of valid inequalities for the G−G m. formulation of the problem. The opposite, i.e., representing flow variables via u_i variables is not possible, though. Therefore, we argue that the one-commodity network flow constraints serve the primal role in the integration of two problem formulations, while the D−L constraints serve the supporting role.

This observation may be useful in practice if introduction of extra variables u_i on top of flow variables to the model is undesirable. On the other hand, eliminating u_i variables (together with corresponding constraints (7.1)) leads to denser linear programming models for a solver to deal with. Few computational experiments indicate that this does not lead to the improved performance of the formulation and increases the solution times.

In order to avoid that, we design another version of the integrated extended mathematical programming formulations where D−L constraints (or, in general, constraints of this type called differently for other problems) are declared as Lazy constraints. Constraints declared as Lazy are originally omitted from the problem formulation and added back only when they are violated. Lazy constraints declared in Gurobi can be of one of three types. We are interested in those where Lazy parameters of the constraints is equal to 3, since such constraints are evaluated for every nonincumbent (fractional) node in the branch-and-bound tree. If a constraint is violated by a fractional node, it is added to the problem formulation, hence strenthening it. Thus, specifying D−L constraints are Lazy ones allows to enjoy the integrated formulation with a stronger LP relaxation while not keeping the majority of the nonviolated D−L constraints in the model all the time. An integrated extended formulation of a routing problem designed in this way will be labled as "lazy" in the section presenting results of computational experiments.

8. Computational Experiments

This section describes the settings, the objective and the results of the computational experiments that have been performed. The main objective of this study is to demonstrate numerically how much stronger (from the LP relaxation standpoint) the integrated formulations are compared to their corresponding components for all the problems we considered in the previous sections. The secondary objective is to demonstrate the efficiency of these simple mixed integer linear programming (MILP) formulations in solving the corresponding problems. However, we argue that extended formulations should not be used as is to solve the problems. Instead, they should be strengthened with the violated inequalities from known families of inequalities at the root node before we proceed to solving the MILP models. For that reason, one or more families of valid inequalities, together with their separation routines, have been introduced to every problem we considered. It also follows that we should not only compare the values of the LP relaxations of the underlying extended problem formulations to those of their integrated versions. Instead, we should be also interested in how much a given formulation can be improved when valid inequalities are introduced and that is why we focus a lot on reporting root relaxation values of different formulations strengthened

with one or several families of valid inequalities. Finally, the approach we use to solve the MILP problems can be called cut-and-branch. While the cut-and-branch approach is not state-of-the-art for either of the considered routing problem, it is arguably the only approach that applies to all the problems we considered in a simple way.

Mathematical programming formulations and cut separation routines have been implemented in a 64-bit Python 3.8.5 environment and solved by Gurobi solver version 11.0.3, Gurobi Optimization, LLC (2020). Some of the formulations and cut separation routines have been implemented using the standard Gurobi interface, while others have been implemented using the specialized Python-MIP interface to Gurobi solver, Santos and Toffolo (2020). That alternative interface allows us to connect to the COIN-OR solver instead of Gurobi without a need to change the code. Problem instances and code behind all experiments can be found in the following repository:

https://github.com/kpavlikov/GG-DL.

Computational experiments were performed using an AMD EPYC 7601 2.20 GHz 16GB RAM machine (4 cores) operated by Windows Server 2019.

Table 1 presents results of computational experiments comparing the objective values of LP relaxations of the corresponding ATSP formulations. A subset of instances from the DIMACS TSP Challenge was used in our set of experiments. It can be observed from the table that the G−G m.+D−L formulation of the problem is stronger than both the G−G m. and D−L formulations, and the magnitude of the improvement is not large, below 1%. The table also provides examples where the G−G m.+D−L formulation can be stronger than the S−D formulation. However, it should be noted that every considered compact formulation can be made equivalent to the D−F−J formulation of the ATSP in the LP relaxation sense by adding all violated D−F−J inequalities.

Tables 2, 3 and 4 display results of computational experiments with the mathematical programming formulations of the ATSP−TW. The problem instances employed in this set of experiments have been considered in López-Ibáñez and Blum (2010) and can be downloaded from https://lopez-ibanez.eu/files/TSPTW/SolomonPotvinBengio.tar.gz. Table 2 provides optimal objective values of LP relaxations of the existing problem formulations together with their improved versions. To illustrate the value of modifications better, the time window parameters tightening and arc elimination procedure outlined in Algorithm 1 has been disabled in this set of experiments. The table demonstrates that the modified versions of the formulations are up to 10% stronger than their original versions. In addition, all four formultions have been strengthened with all violated D−F−J inequalities. While icorporated D−F−J inequalities improve significantly all formulations, and the relative difference between the modified and original formulations noticeably decreases, it is still evident that using the strengthened modified formulations is most recommended. Next, Table 3 fosuses on comparing the modified formulations and their integrated version. Moreover, it incorporates the preprocessing and time window parameter tightening (1) in the analysis. First, LP relaxations of the modified formulations are only slightly stronger than corresponding values reported in Table 2, which may be eplained by the low number of updated time window parameters for most of the considered problem instances. At the same time, it can be noticed that the Y−C−O−S m. formulation can be improved significantly when all cuts are added and preprocessing is executed, see the example of rc_207.3 instance. Most importantly, Table 3 presents LP relaxation values of the the integrated formulation, which is significantly (few per cent points) stronger than either of the components. Incorporation of the violated D−F−J inequalities leads to three different formulations in terms of their strength, with the G−G−E m.+Y−C−O−S m. formulation being unmatched. Table 4 presents results of experiments running the modified and the integrated formulations (strengthened with all violated D−F−J inequalities at the root node and preprocessing and time windows tightenong enabled) as MILP programs for one hour of computer time. The integrated formulation is executed in the "lazy" fashion as well. Experiments marked by the "−" sign indicate that no feasible solution was obtained within the time limit of one hour of running time. For the considered instances, the G−G−E m. formulation, overall, performs best, which can

be explained by the compactness of the formulation, allowing the solver to find feasible solutions easier and prune bnb nodes earlier.

Table 5 contains results of computational experiments with the LP relaxations of the mathematical programming formulations of the ACVRP. The distance matrices and demand values employed to construct the ACVRP instances used in this set of experiments are taken from Fischetti et al. (1994). Every distance matrix and demand values led to three ACVRP instances with different vehicle capacities Q. The number of vehicles is unconstrained in all considered problem instances. The G−G m.+D−L formulation of the problem is considerably stronger than both of its underlying components, with the difference being of a few percent of magnitude. The G−G m.+D−L formulation remains the strongest one even when all problem formulations are strengthened with all violated RC inequalities, even though the D−L formulation provides the same bound in the majority of the cases. That observation is of a special interest since the plain D−L formulation of the problem is considerably weaker than the G−G formulation to begin with for the considered problem instances. Table 6 presents results of solving the mixed integer linear programs of the corresponding formulations. The improved one-commodity network flow formulation G−G m. of the ACVRP performs better than any other formulation, which, again, can be explained by the compactness of the formulation. The largest considered instance, A071-03f with $Q = 500$, was solved to optimality only by the integrated formulation executed in the "lazy" fashion, though.

Tables 7, 8 contain results of computational experiments with the LP relaxations of the ACVRP−TW formulations. The well-known set of problem instances from Solomon (1987) was employed to run the experiments. Table 7 presents results of the experiments with the original formulations of the problem and their modifications, with preprocessing and time window tightening outlined in Algorithm 2 being disabled. Every formulation is strengthened with all violated RC inequalities at the root node, and the computational times (columns named CPU) to obtain those inequalities are reported as well. The table demonstrates both that modifications are highly valuable additions to the problem formulations, resulting is visibly improved quality of LP relaxations, which are amplified significantly with RC cuts. Table 8 reports results of experiments with modified problem formulations where preprocessing, Algorithm 2, is enabled. As in the previous experiments, the integrated formulation of this problem is up to 1% stronger than the original problem formulations. That result holds to problem formulations augmented with violated RC inequalities. Table 9 reports solving ACVRP−TW formulations as mixed integer linear programs. This set of experiments explicitly reveals the main challenge behind the integrated formulation, i.e., its increased size. After one hours of computational time, the integrated formulation has challenges with finding good feasible solutions, resulting in larger optimality gaps reported.

Finally, the remaining tables deal with the experiments solving the HCVRP. Problem instances from Baldacci et al. (2009) are used for that set of experiments. However, in addition to previously considered problem instances with unlimited number of vehicles per vehicle type, this study considers another set of instances, where the maximum number of vehicles of each type is set to a very specific integer number, denoted by μ. When there is no maximum number of existing vehicles provided, this number is created artificially by solving the corresponding bin packing problem with the result denoted by $\bar{\mu}_k$ for every vehicle type k. Table 10 presents results of computational experiments with the LP relaxations of the mathematical programming formulations of the HCVRP. It is remarkable to point out the effect the label-setting problem formulation Y makes on the integrated formulation. Despite the very poor quality of the LP relaxation of this formulation of this problem, it makes a noticeably positive effect on the integrated formulation comparing its bound to the bound of the G−G m. formulation of the problem. Moreover, when formulations are strengthened with violated inequalities from three families, the resulting bound of the integrated formulated matches that of the Y formulation. When it comes to solving the considered problem formulations as mixed integer mathematical programming formulations, no clearly dominating problem formulation appears. Table 11 presents results of such experiments, which indicate that

Instance	G−G		G−G m.		D−L		G−G m.+D−L	
	LP relax.	CPU	LP relax.	CPU	LP relax.	CPU	LP relax.	CPU
d112	11,041.40	0.2	11,041.92	0.2	11,047.13	0.2	11,076.50	0.5
d126	118,716.00	0.3	118,719.00	0.4	121,129.50	0.3	121,167.00	0.8
d134	5,598.13	0.3	5,598.19	0.3	5,604.02	0.3	5,604.77	0.5
d176	8,533.92	0.6	8,533.95	0.7	8,565.51	0.8	8,571.44	1.3
d188	10,142.90	0.4	10,143.19	0.5	10,169.90	0.4	10,191.40	0.8
d563	25,870.70	6.9	25,870.75	8.9	25,923.71	12.3	25,926.50	16.5
d849	37,448.80	19.8	37,448.83	24.0	37,466.00	35.8	37,469.90	106.1
d895	106,963.00	24.7	106,963.14	31.7	107,397.22	25.2	107,403.00	52.5
d932	467,997.00	68.3	467,997.07	78.0	471,547.39	41.9	471,581.00	308.4

Instance	S−D		2CF−ATSP		2CF−ATSP m.		D−F−J		
	LP relax.	CPU	LP relax.	CPU	LP relax.	CPU	LP relax.	# cuts	CPU
d112	11,076.51	0.8	11,075.93	0.2	11,076.45	0.2	11,106.75	41	0.8
d126	121,167.47	1.6	121,165.86	0.3	121,166.84	0.3	123,199.00	129	1.2
d134	5,604.77	1.3	5,604.76	0.4	5,604.77	0.3	5,611.50	43	0.8
d176	8,571.45	2.7	8,571.32	0.7	8,571.37	0.7	8,585.00	82	2.8
d188	10,191.37	2.7	10,191.17	0.9	10,191.28	0.8	10,223.00	53	3.2
d563	25,926.46	50.6	25,926.44	16.2	25,926.45	12.2	25,949.06	122	18.4
d849	37,469.86	217.6	37,469.85	127.6	37,469.85	37.6	37,476.00	34	44.8
d895	107,402.62	158.5	107,402.42	49.0	107,402.44	37.4	107,669.61	401	75.0
d932	471,581.10	1622.6	471,580.64	77.8	471,580.72	88.0	479,837.04	257	32.7

Table 1: Computational comparison of optimal objective values of the LP relaxations of the mixed integer linear ATSP formulations.

Instance	n	G−G−E			G−G−E m.			Y−C−O−S			Y−C−O−S m.		
		LP relax.	+D−F−J	CPU	LP relax.	+D−F−J	CPU	LP relax.	+D−F−J	CPU	LP relax.	+D−F−J	CPU
rc_201.1	20	395.11	410.01	0.0	412.49	425.67	0.0	351.87	384.19	0.0	385.31	399.61	0.0
rc_201.2	26	539.32	588.51	0.0	604.46	649.27	0.0	484.47	533.15	0.0	568.35	579.30	0.0
rc_201.3	32	584.77	677.66	0.1	597.97	726.54	0.1	505.94	572.90	0.1	597.58	618.23	0.1
rc_201.4	26	599.97	690.18	0.1	633.30	736.18	0.0	542.49	611.05	0.0	626.39	671.83	0.0
rc_202.1	33	574.07	615.49	0.1	595.14	637.02	0.1	546.77	601.71	0.1	574.21	613.16	0.1
rc_202.2	14	242.25	304.14	0.0	244.31	304.14	0.0	230.28	304.14	0.0	255.09	304.14	0.0
rc_202.3	29	575.79	642.42	0.1	627.15	735.65	0.1	529.36	605.16	0.0	610.47	651.21	0.0
rc_202.4	28	581.53	622.28	0.1	612.49	667.71	0.1	550.21	598.98	0.0	606.40	625.62	0.1
rc_203.1	19	336.73	409.49	0.0	339.54	421.29	0.0	328.76	406.04	0.0	371.10	410.76	0.0
rc_203.2	23	577.47	662.85	0.1	579.62	670.65	0.1	554.08	656.41	0.0	607.55	666.37	0.1
rc_203.3	37	661.92	710.52	0.2	671.96	718.94	0.2	630.67	697.19	0.0	670.90	708.60	0.1
rc_203.4	15	300.97	314.29	0.0	301.26	314.29	0.0	297.90	314.29	0.0	310.15	314.29	0.0
rc_204.1	46	726.70	805.25	0.3	745.70	807.02	0.3	707.46	804.03	0.1	739.62	805.67	0.2
rc_204.2	33	559.40	608.78	0.1	560.43	608.91	0.1	551.50	608.78	0.0	571.17	608.94	0.1
rc_204.3	24	408.61	439.45	0.0	409.86	439.88	0.0	405.71	439.06	0.0	416.58	439.06	0.0
rc_205.1	14	266.99	270.07	0.0	285.05	291.00	0.0	237.60	255.58	0.0	268.04	274.75	0.0
rc_205.2	27	570.32	605.14	0.0	612.76	661.06	0.0	508.18	540.30	0.0	564.22	592.25	0.0
rc_205.3	35	671.75	736.54	0.1	690.21	755.34	0.1	658.05	728.19	0.1	692.91	740.30	0.1
rc_205.4	28	591.77	643.72	0.1	633.46	670.47	0.0	516.25	605.69	0.0	582.86	632.37	0.0
rc_206.2	37	664.39	699.67	0.1	695.35	731.12	0.1	599.08	645.61	0.1	643.31	663.99	0.1
rc_206.3	25	463.71	489.25	0.0	468.41	496.60	0.0	431.75	458.22	0.0	453.69	472.38	0.0
rc_206.4	38	651.47	704.34	0.1	654.37	714.21	0.1	599.55	668.92	0.0	626.52	678.56	0.1
rc_207.1	34	590.24	631.39	0.1	613.70	648.63	0.1	552.14	622.95	0.0	601.47	628.58	0.1
rc_207.2	31	526.53	568.92	0.1	541.72	576.07	0.1	494.53	562.88	0.0	523.05	564.83	0.0
rc_207.3	33	529.13	562.48	0.1	550.23	578.18	0.1	508.47	556.72	0.0	524.09	559.62	0.1
rc_208.1	38	634.02	691.67	0.1	634.50	691.63	0.2	623.51	689.95	0.0	652.09	689.95	0.1
rc_208.2	29	479.89	524.39	0.1	479.91	524.39	0.1	475.30	523.48	0.0	486.97	523.48	0.0
rc_208.3	36	551.09	588.21	0.1	552.61	588.21	0.1	538.17	588.21	0.0	563.17	588.21	0.1

Table 2: Computational results comparing LP relaxations of the original mixed integer linear ATSP−TW formulations and their improved versions, before and after adding all violated D−F−J inequalities. Preprocessing and time windows tightening according to Algorithm 1 was disabled prior to the experiments.

Instance	$\lvert A_c\rvert$	# new a_i, b_i	**G−G−E m.** LP rel.	+D−F−J	CPU	**Y−C−O−S m.** LP rel.	+D−F−J	CPU	**G−G−E m.+Y−C−O−S m.** LP rel.	+D−F−J	CPU
rc_201.1	127	21	412.49	425.67	0.0	406.13	416.48	0.0	421.26	427.45	0.0
rc_201.2	249	26	604.59	649.37	0.1	644.15	648.57	0.0	646.56	651.04	0.0
rc_201.3	393	34	597.97	726.54	0.1	649.76	712.57	0.1	665.22	728.90	0.1
rc_201.4	249	27	633.30	736.18	0.0	679.66	735.37	0.0	695.83	739.54	0.1
rc_202.1	227	42	595.14	637.02	0.1	584.90	619.87	0.1	614.64	642.24	0.1
rc_202.2	8	21	244.31	304.14	0.0	260.31	304.14	0.0	271.06	304.14	0.0
rc_202.3	274	33	627.15	735.65	0.1	663.92	714.49	0.0	678.94	737.25	0.1
rc_202.4	145	36	612.49	667.71	0.1	625.27	643.53	0.1	651.83	672.91	0.1
rc_203.1	39	27	339.54	421.29	0.0	390.69	422.09	0.0	395.30	422.92	0.0
rc_203.2	115	49	579.62	670.65	0.1	624.82	669.38	0.1	638.93	674.83	0.2
rc_203.3	126	54	671.96	718.94	0.2	677.93	713.77	0.1	694.49	724.83	0.2
rc_203.4	5	25	301.26	314.29	0.0	312.19	314.29	0.0	312.62	314.29	0.0
rc_204.1	57	79	745.70	807.02	0.3	743.18	806.34	0.2	758.49	810.28	0.4
rc_204.2	26	55	560.43	608.91	0.1	571.27	608.94	0.1	577.82	609.42	0.2
rc_204.3	4	42	409.86	439.88	0.0	416.58	439.06	0.0	421.33	439.88	0.1
rc_205.1	44	17	285.05	291.22	0.0	294.67	301.29	0.0	299.62	307.13	0.0
rc_205.2	215	30	612.76	661.06	0.0	638.46	653.38	0.0	651.46	664.76	0.1
rc_205.3	263	42	690.21	755.34	0.1	702.37	749.12	0.1	713.64	758.22	0.2
rc_205.4	240	29	633.46	670.47	0.1	617.22	658.28	0.0	638.57	673.01	0.1
rc_206.2	348	39	695.35	731.12	0.1	699.73	716.27	0.1	714.72	732.99	0.2
rc_206.3	121	27	468.41	496.60	0.0	475.16	488.94	0.0	485.52	497.59	0.0
rc_206.4	362	40	654.37	714.21	0.1	651.57	702.00	0.1	674.30	715.27	0.2
rc_207.1	188	40	613.70	648.63	0.1	622.35	646.19	0.1	631.04	649.57	0.1
rc_207.2	106	36	541.72	576.07	0.1	533.07	569.16	0.1	552.26	577.71	0.1
rc_207.3	107	38	550.23	578.18	0.1	542.36	568.35	0.1	557.24	578.65	0.1
rc_208.1	2	45	634.50	691.63	0.1	652.09	689.95	0.1	655.44	691.80	0.2
rc_208.2	0	40	479.91	524.39	0.1	486.97	523.48	0.0	487.77	524.39	0.1
rc_208.3	0	43	552.61	588.21	0.1	563.17	588.21	0.1	568.98	588.23	0.2

Table 3: Computational results comparing LP relaxations of improved mixed integer linear ATSP−TW formulations with that of their integrated formulation, before and after adding all violated D−F−J inequalities. Preprocessing and time windows tightening according to Algorithm 1 was executed prior to the experiments.

Instance	n	**G−G−E m.** Obj.	Bound	Gap	CPU	**Y−C−O−S m.** Obj.	Bound	Gap	CPU	**G−G−E m.+Y−C−O−S m.** Obj.	Bound	Gap	CPU	**G−G−E m.+Y−C−O−S m.** lazy Obj.	Bound	Gap	CPU
rc_201.1	20	444.54	444.54	0.00	0.0	444.54	444.54	0.00	0.1	444.54	444.54	0.00	0.2	444.54	444.54	0.00	0.2
rc_201.2	26	711.54	711.54	0.00	0.3	711.54	711.54	0.00	0.0	711.54	711.54	0.00	0.4	711.54	711.54	0.00	0.3
rc_201.3	32	790.61	790.61	0.00	2.4	790.61	790.61	0.00	0.5	790.61	790.61	0.00	1.7	790.61	790.61	0.00	1.5
rc_201.4	26	793.64	793.64	0.00	0.3	793.64	793.64	0.00	0.2	793.64	793.64	0.00	0.3	793.64	793.64	0.00	0.6
rc_202.1	33	771.78	756.6	1.97	lim	771.78	738.54	4.31	lim	771.78	722.90	6.33	lim	771.78	729.40	5.49	lim
rc_202.2	14	304.14	304.14	0.00	0.0	304.14	304.14	0.00	0.0	304.14	304.14	0.00	0.0	304.14	304.14	0.00	0.0
rc_202.3	29	837.72	837.72	0.00	2.9	837.72	837.72	0.00	1.5	837.72	837.72	0.00	3.5	837.72	837.72	0.00	2.5
rc_202.4	28	793.03	793.03	0.00	151.9	793.03	793.03	0.00	366.1	793.03	793.03	0.00	119.1	793.03	793.03	0.00	195.2
rc_203.1	19	453.48	453.48	0.00	0.5	453.48	453.48	0.00	0.4	453.48	453.48	0.00	1.5	453.48	453.48	0.00	1.1
rc_203.2	23	784.16	784.16	0.00	483.0	784.16	745.30	4.96	lim	784.16	784.16	0.00	456.7	784.16	784.16	0.00	595.2
rc_203.3	37	817.53	794.39	2.83	lim	817.53	795.76	2.66	lim	817.53	797.49	2.45	lim	817.53	788.49	3.55	lim
rc_203.4	15	314.29	314.29	0.00	0.0	314.29	314.29	0.00	0.0	314.29	314.29	0.00	0.0	314.29	314.29	0.00	0.0
rc_204.1	46	878.64	843.86	3.96	lim	–	830.89	–	lim	–	841.23	–	lim	–	838.89	–	lim
rc_204.2	33	662.16	649.25	1.95	lim	662.16	637.88	3.67	lim	662.16	641.47	3.12	lim	666.52	640.07	3.97	lim
rc_204.3	24	455.03	455.03	0.00	2.6	455.03	455.03	0.00	22.5	455.03	455.03	0.00	3.6	455.03	455.03	0.00	5.8
rc_205.1	14	343.21	343.21	0.00	0.2	343.21	343.21	0.00	0.1	343.21	343.21	0.00	0.2	343.21	343.21	0.00	0.2
rc_205.2	27	755.93	755.93	0.00	7.0	755.93	755.93	0.00	3.3	755.93	755.93	0.00	6.8	755.93	755.93	0.00	9.4
rc_205.3	35	825.06	825.06	0.00	7.1	825.06	825.06	0.00	4.2	825.06	825.06	0.00	4.9	825.06	825.06	0.00	6.8
rc_205.4	28	760.47	760.47	0.00	17.7	760.47	760.47	0.00	5.8	760.47	760.47	0.00	12.0	760.47	760.47	0.00	13.2
rc_206.2	37	828.06	828.06	0.00	308.4	828.06	828.06	0.00	684.8	828.06	828.06	0.00	361.3	828.06	828.06	0.00	364.20
rc_206.3	25	574.42	574.42	0.00	9.3	574.42	574.42	0.00	67.4	574.42	574.42	0.00	13.1	574.42	574.42	0.00	15.9
rc_206.4	38	831.67	818.45	1.59	lim	831.67	805.99	3.09	lim	831.67	831.67	0.00	2033.9	831.67	803.80	3.35	lim
rc_207.1	34	732.68	732.68	0.00	1514.1	732.68	732.68	0.00	1969.7	732.68	732.68	0.00	1098.3	732.68	732.68	0.00	3097.0
rc_207.2	31	701.25	649.40	7.39	lim	–	629.19	–	lim	701.25	643.60	8.22	lim	701.25	653.83	6.76	lim
rc_207.3	33	682.40	653.43	4.25	lim	682.40	620.26	9.11	lim	682.40	654.01	4.16	lim	–	641.34	–	lim
rc_208.1	38	–	733.57	–	lim	–	707.31	–	lim	–	730.97	–	lim	–	730.96	–	lim
rc_208.2	29	533.78	533.78	0.00	4.1	533.78	533.78	0.00	8.1	533.78	533.78	0.00	2.4	533.78	533.78	0.00	6.7
rc_208.3	36	634.44	612.90	3.40	lim	–	597.14	–	lim	644.93	611.65	5.16	lim	657.27	610.64	7.10	lim

Table 4: Computational results of solving the ATSP−TW using mixed integer linear formulations, preprocessed using Algorithm 1 and strengthened with all violated D−F−J inequalities at the root node.

Instance	Q	G−G			G−G m.			D−L			G−G m.+D−L		
		LP relax.	+RCI	CPU	LP relax.	+RCI	CPU	LP relax.	+RCI	CPU	LP relax.	+RCI	CPU
A034-02f	1000	1271.68	1337.00	0.5	1281.40	1337.00	0.6	1195.72	1370.00	0.2	1289.53	1371.15	0.4
	700	1388.46	1503.19	0.6	1404.73	1503.47	1.4	1196.24	1536.00	0.5	1418.49	1536.00	0.8
	500	1564.10	1710.60	1.8	1596.79	1710.60	2.9	1196.99	1736.33	1.0	1630.08	1736.54	0.7
	400	1730.37	1933.59	1.9	1790.05	1933.59	3.9	1197.75	1961.00	1.3	1821.83	1961.19	1.8
A036-03f	1000	1477.11	1605.85	0.8	1485.23	1605.87	1.3	1411.00	1631.90	0.3	1507.55	1631.90	0.8
	750	1585.33	1740.11	1.1	1607.87	1740.11	1.5	1411.00	1757.20	0.5	1623.14	1757.20	0.5
	600	1712.62	1904.00	0.9	1741.66	1904.00	2.3	1411.00	1919.50	1.8	1751.90	1919.50	2.4
	500	1841.21	2020.00	1.8	1893.43	2020.00	2.2	1411.01	2026.31	1.4	1910.92	2026.31	1.3
A039-03f	1000	1566.98	1650.67	1.3	1577.57	1650.67	1.7	1473.70	1654.00	0.2	1596.51	1654.00	0.5
	750	1701.69	1834.50	1.2	1727.72	1834.50	1.6	1474.23	1848.00	0.6	1742.42	1848.00	0.8
	600	1849.40	2023.00	2.3	1893.81	2023.00	3.4	1475.09	2026.75	2.0	1909.76	2026.75	1.9
	500	2005.62	2221.00	2.3	2079.71	2221.00	4.5	1476.61	2225.08	2.2	2095.31	2225.08	2.3
A045-03f	1000	1639.71	1737.50	1.6	1645.58	1737.50	1.7	1573.75	1740.00	0.4	1676.23	1740.00	0.9
	750	1746.14	1890.75	2.0	1768.29	1890.75	3.3	1573.75	1897.11	1.7	1780.06	1897.11	2.9
	600	1888.25	2057.74	3.2	1922.34	2057.74	3.9	1573.76	2071.44	2.8	1932.20	2071.44	3.7
	500	2036.16	2243.00	2.5	2084.31	2243.00	4.1	1573.86	2252.83	3.2	2098.93	2252.83	2.5
A048-03f	1000	1741.59	1837.77	1.0	1743.02	1837.80	1.8	1726.48	1885.00	0.4	1806.70	1885.00	1.2
	750	1803.88	1952.77	2.0	1812.50	1952.77	2.5	1726.86	1975.00	1.0	1866.27	1975.00	1.3
	600	1901.72	2055.00	1.8	1925.43	2055.00	3.0	1727.21	2060.67	1.0	1962.34	2060.67	1.4
	500	2014.27	2223.79	3.9	2068.51	2223.79	6.2	1727.55	2228.71	2.8	2083.57	2228.71	3.8
A056-03f	1000	1537.74	1683.00	1.9	1539.82	1683.00	2.5	1512.53	1717.50	0.5	1601.86	1717.50	1.0
	750	1622.52	1823.25	2.4	1632.49	1823.25	2.8	1513.48	1855.83	1.7	1688.80	1855.83	3.7
	600	1747.59	1988.89	4.3	1778.48	1988.89	7.1	1514.43	2000.22	2.7	1819.52	2000.22	3.7
	500	1889.70	2117.75	6.8	1942.37	2117.75	6.2	1515.41	2126.33	5.8	1980.65	2126.33	4.7
A065-03f	1000	1847.98	1931.92	1.9	1852.83	1931.92	2.8	1761.07	1946.10	1.2	1875.24	1946.10	1.8
	750	1958.99	2098.72	6.9	1977.25	2098.72	6.5	1761.69	2107.33	3.1	2011.27	2107.33	4.3
	600	2104.91	2268.12	6.4	2134.98	2268.12	8.8	1762.68	2275.32	5.1	2168.96	2275.32	7.0
	500	2261.30	2452.00	5.3	2317.88	2452.00	4.8	1764.61	2454.50	4.6	2340.53	2454.50	4.9
A071-03f	1000	1868.92	2002.40	2.3	1871.22	2002.40	4.3	1858.57	2026.00	0.8	1926.07	2026.00	2.0
	750	1952.29	2106.90	4.3	1961.17	2106.90	5.8	1858.60	2139.11	1.8	2006.04	2139.11	2.9
	600	2060.23	2251.50	11.0	2077.34	2251.50	9.0	1858.62	2267.68	5.3	2123.20	2267.68	9.9
	500	2176.96	2354.00	13.4	2208.01	2354.00	17.2	1858.91	2358.06	6.4	2252.63	2358.06	14.3

Table 5: Computational results comparing LP relaxations of mixed integer linear ACVRP formulations, before and after adding all violated RC inequalities.

9. Conclusion

This study considered two fundamental approaches to formulate the TSP and many versions of the VRP with help of extra variables. Despite the different underlying ideas used to develop these approaches, it turns out that both approaches can provide equivalent information about the elapsed route time, position, or the cumulative load of a vehicle at every node. Therefore, a problem formulation can be obtained where variables and constraints are introduced for two approaches simultaneously. The integration of the approaches becomes meaningful once we equate the expressions representing the quantity of interest (elapsed time or vehicle load) that appear from both approaches. The resulting mathematical programming formulation of a problem is demonstrated to have considerably stronger linear programming relaxation. A computational study demonstrates the improved quality of lower bounds that can be obtained at the root node of the integrated problem formulation compared to that of the underlying extended problem formulations, even when every formulation is strengthened with help of problem-specific valid inequalities.

10. Data Availability

All datasets involved in this study are publicly available referenced to in Section 8.

11. Declarations

The author has no relevant financial or non-financial interests to disclose.

Instance	Q	G–G m.				D–L				G–G m.+D–L				G–G m.+D–L lazy			
		Obj.	Bound	Gap	CPU	Obj.	Bound	Gap	CPU	Obj.	Bound	Gap	CPU	Obj.	Bound	Gap	CPU
A034-02f	1000	1406	1406	0.00	0.9	1406	1406	0.00	0.8	1406	1406	0.00	1.7	1406	1406	0.00	7.0
	700	1553	1553	0.00	0.7	1553	1553	0.00	0.5	1553	1553	0.00	2.8	1553	1553	0.00	4.0
	500	1773	1773	0.00	2.3	1773	1773	0.00	2.0	1773	1773	0.00	2.2	1773	1773	0.00	19.0
	400	2016	2016	0.00	4.3	2016	2016	0.00	1.9	2016	2016	0.00	9.8	2016	2016	0.00	17.5
A036-03f	1000	1644	1644	0.00	0.4	1644	1644	0.00	0.3	1644	1644	0.00	0.6	1644	1644	0.00	4.3
	750	1774	1774	0.00	1.3	1774	1774	0.00	0.5	1774	1774	0.00	1.4	1774	1774	0.00	5.0
	600	1966	1966	0.00	2.5	1966	1966	0.00	1.6	1966	1966	0.00	4.5	1966	1966	0.00	15.7
	500	2110	2110	0.00	3.6	2110	2110	0.00	5.3	2110	2110	0.00	16.9	2110	2110	0.00	40.0
A039-03f	1000	1654	1654	0.00	0.3	1654	1654	0.00	0.0	1654	1654	0.00	0.1	1654	1654	0.00	0.3
	750	1896	1896	0.00	6.6	1896	1896	0.00	6.0	1896	1896	0.00	3.9	1896	1896	0.00	24.8
	600	2053	2053	0.00	1.2	2053	2053	0.00	2.2	2053	2053	0.00	2.9	2053	2053	0.00	17.6
	500	2289	2289	0.00	5.3	2289	2289	0.00	8.8	2289	2289	0.00	19.9	2289	2289	0.00	21.5
A045-03f	1000	1740	1740	0.00	0.5	1740	1740	0.00	0.1	1740	1740	0.00	0.5	1740	1740	0.00	1.4
	750	1907	1907	0.00	1.3	1907	1907	0.00	0.7	1907	1907	0.00	1.3	1907	1907	0.00	4.4
	600	2096	2096	0.00	1.9	2096	2096	0.00	2.8	2096	2096	0.00	3.9	2096	2096	0.00	7.8
	500	2303	2303	0.00	10.7	2303	2303	0.00	4.2	2303	2303	0.00	24.4	2303	2303	0.00	8.0
A048-03f	1000	1891	1891	0.00	0.5	1891	1891	0.00	0.3	1891	1891	0.00	0.7	1891	1891	0.00	5.8
	750	2036	2036	0.00	9.4	2036	2036	0.00	14.1	2036	2036	0.00	9.7	2036	2036	0.00	23.3
	600	2099	2099	0.00	3.7	2099	2099	0.00	3.3	2099	2099	0.00	5.1	2099	2099	0.00	24.7
	500	2283	2283	0.00	51.5	2283	2283	0.00	24.3	2283	2283	0.00	32.7	2283	2283	0.00	99.8
A056-03f	1000	1739	1739	0.00	2.1	1739	1739	0.00	1.8	1739	1739	0.00	3.2	1739	1739	0.00	9.7
	750	1906	1906	0.00	17.5	1906	1906	0.00	96.0	1906	1906	0.00	41.1	1906	1906	0.00	116.5
	600	2007	2007	0.00	2.9	2007	2007	0.00	0.8	2007	2007	0.00	2.5	2007	2007	0.00	6.6
	500	2165	2165	0.00	374.0	2165	2165	0.00	622.3	2165	2165	0.00	308.6	2165	2165	0.00	281.2
A065-03f	1000	1974	1974	0.00	2.5	1974	1974	0.00	1.9	1974	1974	0.00	4.4	1974	1974	0.00	33.8
	750	2173	2173	0.00	116.9	2173	2173	0.00	297.2	2173	2173	0.00	142.4	2173	2173	0.00	812.9
	600	2373	2347	1.10	lim	2382	2316	2.77	lim	2387	2329	2.43	lim	2399	2334	2.71	lim
	500	2567	2531	1.40	lim	2567	2525	1.64	lim	2571	2518	2.06	lim	2575	2503	2.80	lim
A071-03f	1000	2054	2054	0.00	2.3	2054	2054	0.00	0.9	2054	2054	0.00	1.5	2054	2054	0.00	15.7
	750	2209	2209	0.00	99.9	2209	2209	0.00	1482.5	2209	2209	0.00	248.5	2209	2209	0.00	326.8
	600	2356	2347	0.38	lim	2356	2332	1.02	lim	2367	2329	1.61	lim	2385	2322	2.64	lim
	500	2457	2438	0.77	lim	2457	2430	1.10	lim	2457	2445	0.49	lim	2457	2457	0.00	3110.9

Table 6: Computational results comparing mixed integer linear ACVRP formulations, strengthened with all violated RC inequalities at the root node.

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Inst.	G−G−E			G−G−E m.			Y−C−O−S			Y−C−O−S m.		
	LP rel.	+RCI	CPU	LP rel.	+RCI	CPU	LP rel.	+RCI	CPU	LP rel.	+RCI	CPU
C101	750.49	816.69	17.2	825.04	827.30	0.6	754.07	820.63	21.1	784.27	825.95	39.8
C102	750.49	816.69	22.4	795.42	824.82	21.7	751.27	818.20	18.5	769.28	822.81	22.3
C103	750.49	816.69	20.8	769.17	823.71	40.1	750.91	817.75	23.8	752.83	819.22	27.9
C104	750.49	816.69	18.0	754.80	819.10	37.8	750.49	816.77	17.7	750.51	817.13	21.6
C105	750.49	816.69	16.8	824.84	827.30	1.2	750.90	818.75	21.7	764.23	822.63	21.7
C106	750.49	816.69	17.2	809.32	827.30	8.1	751.95	818.04	21.1	764.02	822.36	33.2
C107	750.49	816.69	21.6	824.84	827.30	1.0	750.52	817.92	26.7	754.97	819.99	30.1
C108	750.49	816.69	21.8	785.17	824.34	44.6	750.49	817.38	22.7	751.76	818.61	22.1
C109	750.49	816.69	3.7	772.45	822.31	36.2	750.49	816.69	21.9	750.49	817.02	24.0
C201	549.60	578.60	7.6	589.10	589.10	0.0	555.89	583.64	4.9	569.77	586.03	4.8
C202	549.60	578.60	4.1	583.23	588.99	4.2	551.10	581.49	4.3	557.87	584.54	4.2
C203	549.60	578.60	3.9	569.36	582.90	8.2	549.77	579.58	6.0	550.49	580.65	5.5
C204	549.60	578.60	3.5	565.09	580.92	9.2	549.60	578.91	6.7	549.88	579.28	4.0
C205	549.60	578.60	3.8	580.86	585.73	8.6	551.11	581.40	3.6	557.14	584.00	4.0
C206	549.60	578.60	3.5	578.51	585.48	10.3	549.96	580.27	3.1	553.66	582.83	5.0
C207	549.60	578.60	3.7	574.31	584.88	11.5	550.88	580.55	4.1	556.84	583.24	6.7
C208	549.60	578.60	4.2	573.72	585.26	10.5	549.60	579.69	5.5	551.73	581.75	4.3
R101	769.89	801.03	34.6	1619.04	1619.04	0.1	825.27	847.54	18.3	1440.78	1440.78	0.1
R102	769.89	801.03	30.3	1275.01	1276.58	0.6	802.38	827.42	21.9	1014.02	1034.03	8.1
R103	769.89	801.03	38.9	1006.94	1009.32	3.5	781.04	810.40	27.9	825.46	850.99	20.4
R104	769.89	801.03	35.4	846.38	851.87	17.9	770.96	801.21	29.9	776.14	806.50	41.0
R105	769.89	801.03	31.4	1276.86	1277.05	0.3	774.93	803.19	23.1	984.57	993.67	4.3
R106	769.89	801.03	32.9	1072.29	1072.79	1.7	772.28	801.52	30.4	848.84	875.30	20.6
R107	769.89	801.03	32.6	917.53	920.11	4.9	771.04	801.42	29.6	787.89	816.02	26.8
R108	769.89	801.03	30.8	831.18	838.61	20.5	769.99	801.03	35.1	771.44	801.72	36.4
R109	769.89	801.03	38.4	1014.40	1014.50	0.6	769.89	801.03	39.4	790.32	818.68	24.1
R110	769.89	801.03	33.1	924.00	928.62	2.8	769.89	801.03	36.2	772.06	803.19	40.8
R111	769.89	801.03	34.1	907.07	913.11	4.3	770.18	801.03	35.8	775.06	805.69	33.5
R112	769.89	801.03	40.6	850.95	859.17	7.8	769.89	801.03	37.6	769.89	801.03	43.6
R201	629.15	644.24	7.6	1052.07	1060.06	2.4	642.93	653.53	3.7	873.96	881.20	1.5
R202	629.15	644.24	7.4	851.82	855.71	0.8	636.55	648.35	3.8	709.15	717.46	2.6
R203	629.15	644.24	7.5	701.84	705.56	5.2	632.52	646.67	5.0	648.44	660.50	6.3
R204	629.15	644.24	7.3	642.78	650.61	16.9	629.46	644.26	9.2	630.25	645.05	6.6
R205	629.15	644.24	9.6	820.00	827.74	3.7	630.85	645.32	4.5	683.99	693.73	4.0
R206	629.15	644.24	9.5	719.11	725.00	3.6	629.76	644.49	6.5	647.54	659.26	6.9
R207	629.15	644.24	8.4	665.69	672.90	6.5	629.36	644.34	6.6	631.92	647.45	6.0
R208	629.15	644.24	7.9	637.00	645.19	8.9	629.16	644.24	8.4	629.17	644.25	7.8
R209	629.15	644.24	7.5	713.08	717.89	7.4	629.21	644.28	6.7	643.65	654.66	4.7
R210	629.15	644.24	7.2	709.79	718.76	5.1	630.36	644.62	6.6	644.88	656.86	5.4
R211	629.15	644.24	9.0	642.30	649.10	9.4	629.15	644.24	7.6	629.15	644.24	9.5
RC101	886.10	961.70	65.4	1460.78	1471.87	1.2	890.14	964.34	54.7	1093.03	1129.48	16.6
RC102	886.10	961.70	55.8	1195.40	1207.72	8.2	888.27	963.35	62.9	940.30	1001.19	48.2
RC103	886.10	961.70	59.3	1022.47	1057.31	21.9	886.49	962.07	77.0	898.01	970.77	68.5
RC104	886.10	961.70	56.7	941.06	996.68	41.8	886.34	961.70	80.4	890.50	964.40	83.2
RC105	886.10	961.70	57.1	1250.22	1264.64	6.7	889.24	963.09	60.7	970.76	1028.42	30.6
RC106	886.10	961.70	66.6	1144.73	1160.79	8.8	886.10	961.70	77.0	903.76	974.27	51.9
RC107	886.10	961.70	55.1	1015.03	1046.28	20.3	886.10	961.70	87.0	887.21	962.32	80.8
RC108	886.10	961.70	64.0	961.43	999.69	27.5	886.10	961.70	99.9	886.10	961.70	83.0
RC201	591.41	651.65	8.1	1061.83	1126.41	7.6	606.10	663.44	5.0	842.48	880.23	4.3
RC202	591.41	651.65	7.2	796.02	822.27	9.8	597.54	656.73	7.5	661.03	701.05	6.0
RC203	591.41	651.65	7.6	661.82	703.10	17.3	592.84	653.22	7.8	603.74	660.52	6.3
RC204	591.41	651.65	6.6	606.34	658.30	17.5	592.53	652.23	9.0	594.13	653.50	8.2
RC205	591.41	651.65	7.0	886.13	914.06	7.1	597.03	656.02	7.0	716.12	750.10	5.1
RC206	591.41	651.65	7.4	822.90	858.66	13.2	592.42	652.69	6.7	650.54	695.92	5.0
RC207	591.41	651.65	9.8	698.70	731.11	15.5	591.64	651.83	6.8	601.39	658.65	6.6
RC208	591.41	651.65	7.5	609.38	657.39	25.7	591.41	651.65	8.9	591.41	651.65	10.0

Table 7: Linear programming relaxations of the original extended ACVRP−TW formulations. Time windows parameter tightening and arc elimination outlined in Algorithm 2 is disabled.

Inst.	$\lvert A_e \rvert$	# new a_i, b_i	**G−G−E m.**			**Y−C−O−S m.**			**G−G−E m.+Y−C−O−S m.**		
			LP relax.	+RCI	CPU	LP relax.	+RCI	CPU	LP relax.	+RCI	CPU
C101	5592	108	825.04	827.30	0.4	825.01	827.30	0.2	825.04	827.30	0.3
C102	3523	131	795.42	824.82	12.0	792.00	824.65	10.6	795.54	824.84	10.2
C103	1734	152	769.17	823.71	34.8	756.38	820.94	16.1	769.57	823.72	42.9
C104	577	177	754.80	819.10	40.1	750.92	817.70	18.3	755.21	819.13	57.2
C105	5051	108	824.84	827.30	0.6	824.84	827.30	0.6	824.84	827.30	0.7
C106	4684	111	809.32	827.30	5.6	785.61	827.30	2.4	810.28	827.30	6.4
C107	4485	113	824.84	827.30	0.6	824.84	827.30	1.2	824.84	827.30	0.6
C108	3761	115	785.17	824.34	39.1	761.71	822.70	19.3	789.97	825.65	25.5
C109	2646	117	772.45	822.31	36.8	752.87	819.75	18.1	774.13	823.56	33.8
C201	4880	102	589.10	589.10	0.0	589.10	589.10	0.0	589.10	589.10	0.0
C202	2751	127	583.23	588.99	1.8	571.09	588.88	2.2	583.26	588.99	1.9
C203	1234	152	569.36	582.90	6.1	557.45	582.54	2.4	569.93	582.90	8.3
C204	311	176	565.09	580.92	9.3	552.60	580.02	3.2	565.81	580.94	17.5
C205	4403	105	580.86	585.73	3.9	567.54	585.15	2.5	582.62	586.06	3.3
C206	3888	106	578.51	585.48	3.9	564.20	584.50	3.7	579.88	585.95	5.4
C207	3522	106	574.31	584.88	4.4	562.46	584.69	4.7	575.27	584.98	6.0
C208	3436	109	573.72	585.26	5.5	559.08	584.20	3.7	574.74	585.69	6.6
R101	6869	106	1619.06	1619.06	0.1	1620.72	1620.72	0.0	1620.86	1620.86	0.0
R102	4512	128	1275.23	1276.63	0.5	1124.94	1141.07	3.6	1281.44	1282.57	0.3
R103	2764	152	1006.94	1009.32	2.1	879.36	903.50	14.7	1013.42	1015.45	1.5
R104	1198	176	846.38	851.87	9.2	782.25	812.34	25.6	847.71	852.93	6.8
R105	5858	107	1276.86	1277.05	0.2	1207.72	1210.08	0.6	1284.85	1284.88	0.3
R106	3711	131	1072.31	1072.89	0.9	945.85	971.95	10.5	1081.04	1081.26	0.7
R107	2186	155	917.53	920.11	2.1	814.85	839.92	20.4	921.22	923.51	1.9
R108	898	176	831.18	838.61	11.1	775.90	806.13	70.6	832.27	839.56	14.3
R109	4086	113	1015.32	1015.41	0.4	856.50	882.49	9.1	1023.41	1023.88	0.4
R110	2148	121	924.00	928.62	2.4	782.93	811.87	17.3	928.16	931.78	1.9
R111	2153	126	907.07	913.11	4.0	798.05	827.48	18.6	911.96	916.56	3.0
R112	90	131	850.95	859.17	8.3	769.89	801.03	40.0	853.37	860.70	6.8
R201	4183	108	1052.07	1060.06	1.4	1052.69	1062.50	1.0	1063.71	1071.21	1.6
R202	2382	132	851.82	855.71	0.8	784.43	792.30	2.9	865.42	867.12	0.8
R203	1050	156	701.84	705.56	3.8	667.64	676.80	3.4	707.88	710.45	4.4
R204	255	176	642.78	650.61	14.1	632.97	646.95	9.6	644.78	653.07	21.6
R205	2774	111	820.00	827.74	5.4	786.94	797.61	2.4	824.14	831.96	5.0
R206	1580	133	719.11	725.00	3.0	686.18	696.48	4.1	723.27	729.42	4.2
R207	723	157	665.69	672.90	9.8	650.18	662.93	5.9	667.57	675.24	13.0
R208	164	176	637.00	645.19	16.7	629.72	644.25	9.1	637.66	646.07	19.6
R209	1584	117	713.08	717.89	3.4	683.54	691.60	3.4	717.14	721.90	6.8
R210	1535	120	709.79	718.76	4.5	681.83	693.40	4.9	715.64	723.84	3.5
R211	104	124	642.30	649.10	10.2	629.15	644.24	9.0	642.72	649.69	6.9
RC101	6468	107	1460.78	1471.87	1.2	1294.10	1342.95	4.1	1476.48	1486.51	1.0
RC102	4478	131	1195.40	1207.72	3.8	1017.72	1071.16	22.4	1205.95	1218.28	4.0
RC103	2775	155	1022.47	1057.31	14.3	915.63	987.79	41.6	1027.03	1062.60	17.9
RC104	1219	176	941.06	996.68	37.6	896.81	969.11	52.7	945.64	1001.22	45.7
RC105	4871	120	1251.93	1266.02	4.1	1079.79	1123.02	14.3	1260.76	1273.81	2.9
RC106	4731	114	1144.73	1160.79	6.8	981.31	1044.27	22.5	1150.55	1166.57	5.9
RC107	2572	123	1016.40	1047.51	18.6	899.58	972.92	42.7	1022.78	1052.60	18.3
RC108	797	129	961.43	996.69	29.2	886.10	961.71	82.7	962.56	1000.85	32.9
RC201	4187	106	1067.76	1132.42	4.0	1062.42	1126.51	2.4	1085.41	1143.96	3.9
RC202	2350	130	801.22	827.85	6.3	735.53	763.48	5.2	815.84	841.81	8.8
RC203	1044	154	661.82	703.10	13.4	627.11	673.72	5.9	668.00	707.64	16.3
RC204	246	176	606.34	658.30	16.4	597.68	655.07	6.9	609.06	661.07	25.8
RC205	2928	115	886.13	914.06	4.2	838.41	862.81	4.3	894.88	921.76	5.9
RC206	2735	109	822.90	858.66	10.6	789.21	829.29	3.8	830.57	864.26	7.7
RC207	1484	117	698.70	731.11	15.5	648.19	691.31	13.1	703.87	732.78	13.9
RC208	9	123	609.38	657.39	23.3	591.41	651.65	8.3	610.16	658.04	14.9

Table 8: Linear programming relaxations of the most advanced extended ACVRP−TW formulations. Time windows parameter tightening and arc elimination outlined in Algorithm 2 is enabled.

Inst.	G–G–E m.				Y–C–O–S m.				G–G–E m.+Y–C–O–S m.				G–G–E m.+Y–C–O–S m. lazy			
	Obj.	Bound	Gap	CPU	Obj.	Bound	Gap	CPU	Obj.	Bound	Gap	CPU	Obj.	Bound	Gap	CPU
C101	827.30	827.30	0.00	1.5	827.30	827.30	0.00	0.4	827.30	827.30	0.00	0.6	827.30	827.30	0.00	0.6
C102	827.30	827.30	0.00	9.0	827.30	827.30	0.00	2.5	827.30	827.30	0.00	2.6	827.30	827.30	0.00	6.1
C103	826.30	826.30	0.00	17.2	826.30	826.30	0.00	11.3	826.30	826.30	0.00	15.3	826.30	826.30	0.00	33.0
C104	822.90	822.90	0.00	48.5	822.90	822.90	0.00	26.7	822.90	822.90	0.00	57.2	822.90	822.90	0.00	91.5
C105	827.30	827.30	0.00	1.7	827.30	827.30	0.00	0.7	827.30	827.30	0.00	1.0	827.30	827.30	0.00	1.0
C106	827.30	827.30	0.00	2.8	827.30	827.30	0.00	1.2	827.30	827.30	0.00	1.2	827.30	827.30	0.00	2.0
C107	827.30	827.30	0.00	2.1	827.30	827.30	0.00	0.9	827.30	827.30	0.00	1.1	827.30	827.30	0.00	1.4
C108	827.30	827.30	0.00	8.6	827.30	827.30	0.00	6.1	827.30	827.30	0.00	5.9	827.30	827.30	0.00	7.1
C109	827.30	827.30	0.00	22.0	827.30	827.30	0.00	18.1	827.30	827.30	0.00	15.9	827.30	827.30	0.00	11.9
C201	589.10	589.10	0.00	2.1	589.10	589.10	0.00	0.4	589.10	589.10	0.00	0.9	589.10	589.10	0.00	1.0
C202	589.10	589.10	0.00	6.3	589.10	589.10	0.00	1.5	589.10	589.10	0.00	3.3	589.10	589.10	0.00	3.7
C203	588.70	588.70	0.00	16.5	588.70	588.70	0.00	10.7	588.70	588.70	0.00	17.9	588.70	588.70	0.00	19.4
C204	588.10	588.10	0.00	20.6	588.10	588.10	0.00	18.1	588.10	588.10	0.00	20.0	588.10	588.10	0.00	31.7
C205	586.40	586.40	0.00	5.2	586.40	586.40	0.00	1.7	586.40	586.40	0.00	3.9	586.40	586.40	0.00	3.6
C206	586.00	586.00	0.00	4.8	586.00	586.00	0.00	2.6	586.00	586.00	0.00	2.6	586.00	586.00	0.00	4.9
C207	585.80	585.80	0.00	5.2	585.80	585.80	0.00	3.3	585.80	585.80	0.00	4.0	585.80	585.80	0.00	16.1
C208	585.80	585.80	0.00	8.2	585.80	585.80	0.00	3.1	585.80	585.80	0.00	4.8	585.80	585.80	0.00	13.8
R101	1637.70	1637.70	0.00	1.9	1637.70	1637.70	0.00	0.7	1637.70	1637.70	0.00	0.7	1637.70	1637.70	0.00	0.7
R102	1468.50	1384.50	5.72	lim	1472.10	1302.50	11.52	lim	1466.60	1392.30	5.07	lim	1468.50	1399.40	4.71	lim
R103	1208.70	1102.10	8.82	lim	1220.80	1012.80	17.04	lim	1217.20	1115.30	8.37	lim	1208.70	1116.70	7.61	lim
R104	1016.40	886.30	12.80	lim	997.70	835.40	16.27	lim	996.10	888.20	10.83	lim	1021.80	893.40	12.57	lim
R105	1355.30	1355.30	0.00	351.1	1355.30	1355.30	0.00	461.5	1355.30	1355.30	0.00	154.7	1355.30	1355.30	0.00	174.5
R106	1257.00	1159.30	7.77	lim	1251.90	1068.20	14.67	lim	1238.90	1166.00	5.88	lim	1255.10	1174.20	6.45	lim
R107	1077.60	987.00	8.41	lim	1080.00	908.70	15.86	lim	1084.10	998.40	7.91	lim	1255.10	1174.20	6.45	lim
R108	972.00	865.10	11.00	lim	961.80	820.80	14.66	lim	957.70	866.90	9.48	lim	990.00	871.80	11.94	lim
R109	1147.20	1100.20	4.10	lim	1148.00	983.40	14.34	lim	1146.90	1105.40	3.62	lim	1146.90	1106.00	3.57	lim
R110	1076.90	991.50	7.93	lim	1145.80	859.90	24.95	lim	1068.00	997.80	6.57	lim	1108.50	1001.80	9.63	lim
R111	1048.70	975.40	6.99	lim	1054.80	883.70	16.22	lim	1049.00	982.40	6.35	lim	1055.80	982.40	6.95	lim
R112	953.90	888.60	6.85	lim	1012.20	804.30	20.54	lim	970.80	888.70	8.46	lim	976.30	891.30	8.71	lim
R201	1143.20	1143.20	0.00	73.6	1143.20	1143.20	0.00	22.9	1143.20	1143.20	0.00	31.7	1143.20	1143.20	0.00	73.9
R202	1040.30	943.70	9.29	lim	1045.20	879.50	15.85	lim	1030.10	959.50	6.85	lim	1064.90	960.70	9.78	lim
R203	891.80	769.50	13.71	lim	924.10	718.70	22.23	lim	877.30	779.30	11.17	lim	890.40	782.20	12.15	lim
R204	733.50	664.40	9.42	lim	746.80	654.60	12.35	lim	737.90	668.00	9.47	lim	772.50	667.80	13.55	lim
R205	957.80	898.50	6.19	lim	966.70	890.10	7.92	lim	954.70	904.40	5.27	lim	988.10	899.00	9.02	lim
R206	908.30	792.10	12.79	lim	909.50	766.00	15.78	lim	897.80	797.40	11.18	lim	955.30	797.70	16.50	lim
R207	813.70	715.10	12.12	lim	819.00	694.70	15.18	lim	812.10	719.30	11.43	lim	841.10	721.60	14.21	lim
R208	722.80	653.90	9.53	lim	721.80	650.00	9.95	lim	717.60	657.20	8.42	lim	720.10	657.90	8.64	lim
R209	890.20	787.80	11.50	lim	886.80	764.30	13.81	lim	888.50	788.70	11.23	lim	898.70	790.80	12.01	lim
R210	925.80	789.90	14.68	lim	913.60	769.20	15.81	lim	914.90	799.30	12.64	lim	962.40	801.00	16.77	lim
R211	760.40	664.00	12.68	lim	799.00	648.80	18.80	lim	778.00	665.70	14.43	lim	801.00	666.40	16.80	lim
RC101	1619.80	1619.80	0.00	701.8	1619.80	1619.80	0.00	210.3	1619.80	1619.80	0.00	187.2	1619.80	1619.80	0.00	144.8
RC102	1476.70	1327.90	10.08	lim	1476.10	1218.70	17.44	lim	1478.90	1349.70	8.74	lim	1477.70	1349.50	8.68	lim
RC103	1300.20	1136.30	12.61	lim	1329.30	1042.10	21.61	lim	1328.90	1145.40	13.81	lim	1273.60	1147.40	9.91	lim
RC104	1154.30	1032.70	10.53	lim	1179.10	984.10	16.54	lim	1154.40	1034.50	10.39	lim	1178.10	1041.30	11.61	lim
RC105	1525.00	1409.90	7.55	lim	1514.30	1306.20	13.74	lim	1544.50	1428.10	7.54	lim	1514.20	1432.60	5.39	lim
RC106	1380.60	1263.30	8.50	lim	1404.30	1122.90	20.04	lim	1375.40	1270.00	7.66	lim	1373.90	1271.70	7.44	lim
RC107	1209.30	1107.00	8.46	lim	1340.80	1003.10	25.19	lim	1210.30	1116.00	7.79	lim	1258.50	1120.20	10.99	lim
RC108	1132.50	1025.70	9.43	lim	1131.70	965.80	14.66	lim	1144.80	1031.30	9.91	lim	1142.2	1036.40	9.26	lim
RC201	1261.80	1261.80	0.00	142.8	1261.80	1261.80	0.00	32.7	1261.80	1261.80	0.00	80.0	1261.80	1261.80	0.00	108.4
RC202	1096.30	953.60	13.02	lim	1110.00	886.70	20.12	lim	1095.80	968.80	11.59	lim	1119.40	977.00	12.72	lim
RC203	976.20	752.70	22.89	lim	964.00	722.30	25.07	lim	971.70	764.00	21.37	lim	1028.10	766.20	25.47	lim
RC204	798.00	674.90	15.43	lim	822.00	668.90	18.63	lim	789.50	681.70	13.65	lim	808.30	683.40	15.45	lim
RC205	1157.70	1043.00	9.91	lim	1157.70	1049.80	9.32	lim	1164.90	1046.30	10.18	lim	1169.80	1052.00	10.07	lim
RC206	1075.20	960.90	10.63	lim	1066.90	943.70	11.55	lim	1070.70	964.10	9.96	lim	1071.90	966.90	9.80	lim
RC207	1010.20	802.00	20.61	lim	962.90	769.80	20.05	lim	974.00	806.90	17.16	lim	1106.40	808.60	26.92	lim
RC208	833.60	667.30	19.95	lim	802.10	654.50	18.40	lim	783.40	670.80	14.37	lim	795.70	670.20	15.77	lim

Table 9: Computational results of solving the ACVRP−TW instances using considered mixed integer linear programming formulations, preprocessed using Algorithm 2 and strengthened with all violated RC inequalities at the root node.

Instance	n	μ	G–G			G–G m.			Y			G–G m.+Y		
			LP relax.	+cuts	CPU	LP relax.	+cuts	CPU	LP relax.	+cuts	CPU	LP relax.	+cuts	CPU
3	20	2	886.73	925.90	0.6	889.53	925.90	0.5	227.78	925.90	0.4	892.51	925.92	0.6
		$\bar{\mu}_k$	848.03	916.56	0.6	852.75	916.86	0.6	227.78	915.00	0.4	856.55	917.47	0.7
4	20	2	6769.39	7289.50	1.0	6771.50	7289.50	0.2	225.31	7289.50	0.2	6776.15	7289.50	0.4
		$\bar{\mu}_k$	6220.76	6404.50	1.5	6222.49	6404.50	1.5	225.31	6404.50	1.3	6229.29	6404.50	2.1
5	20	2	887.96	963.27	1.1	897.98	963.38	0.5	227.78	963.00	0.6	903.49	963.70	0.7
		$\bar{\mu}_k$	870.20	963.56	2.0	883.78	963.38	0.7	227.78	963.00	0.6	885.45	963.70	1.0
6	20	3	6564.46	6909.50	2.4	6568.70	6909.50	0.7	225.31	6909.50	0.9	6576.86	6909.50	1.3
		$\bar{\mu}_k$	6221.68	6494.15	5.4	6231.43	6494.15	3.1	225.31	6494.15	1.9	6238.38	6494.15	2.6
13	50	3	2279.29	2337.52	10.3	2281.55	2338.34	7.5	364.49	2337.44	11.5	2282.13	2337.44	11.5
		$\bar{\mu}_k$	2204.54	2327.95	6.7	2205.91	2327.95	5.8	364.49	2327.95	7.6	2209.44	2327.95	11.3
14	50	3	9504.86	9993.50	9.4	9505.98	9993.44	6.0	369.44	9987.02	3.4	9515.87	9987.02	7.8
		$\bar{\mu}_k$	8596.70	9053.96	9.1	8597.59	9053.96	6.7	369.44	9053.96	5.3	8610.05	9053.96	10.7
15	50	3	2486.84	2581.29	10.2	2492.35	2574.17	3.8	393.84	2574.42	4.7	2499.18	2574.18	6.3
		$\bar{\mu}_k$	2273.34	2483.98	11.6	2274.91	2485.84	14.2	393.84	2480.45	9.2	2303.63	2480.92	16.2
16	50	4	2615.00	2673.36	5.2	2621.65	2673.93	4.8	395.48	2673.89	6.7	2628.18	2673.37	8.6
		$\bar{\mu}_k$	2518.84	2638.80	7.9	2522.62	2638.80	9.0	395.48	2638.80	8.3	2529.55	2638.90	14.0
17	75	4	1571.63	1688.15	36.0	1573.07	1686.64	37.7	488.70	1686.85	32.1	1578.22	1685.99	63.4
		$\bar{\mu}_k$	1557.73	1674.64	35.3	1558.95	1674.22	50.6	488.70	1674.30	35.7	1565.20	1674.34	60.4
18	75	3	2300.32	2426.21	25.4	2301.17	2426.53	33.3	487.23	2426.73	23.0	2304.83	2426.60	67.3
		$\bar{\mu}_k$	2024.15	2245.31	39.8	2024.50	2245.31	49.6	487.23	2245.31	29.4	2030.40	2245.31	107.5
19	100	4	9330.67	9718.75	58.7	9332.17	9718.75	93.9	591.51	9718.75	34.5	9347.43	9718.75	140.6
		$\bar{\mu}_k$	8143.01	8586.21	333.6	8144.23	8586.21	404.5	591.51	8586.21	162.0	8175.07	8586.21	959.40
20	100	4	4037.47	4131.69	86.1	4043.50	4126.53	98.7	588.94	4125.27	39.8	4053.82	4123.30	164.9
		$\bar{\mu}_k$	3584.81	3850.14	1038.1	3586.49	3848.35	1161.9	588.94	3851.44	627.9	3646.40	3847.98	1669.6

Table 10: Computational results comparing LP relaxations of MILP HCVRP formulations, before and after adding all violated inequalities from three families.

Inst.	n	μ	G–G m.				Y				G–G m.+Y				G–G m.+Y lazy			
			Obj.	Bound	Gap	CPU	Obj.	Bound	Gap	CPU	Obj.	Bound	Gap	CPU	Obj.	Bound	Gap	CPU
3	20	2	960	960	0.00	114.9	960	960	0.00	50.0	960	960	0.00	80.9	960	960	0.00	421.5
		$\bar{\mu}_k$	960	960	0.00	484.8	960	960	0.00	338.1	960	960	0.00	329.5	960	960	0.00	2393.3
4	20	2	7300	7300	0.00	3.1	7300	7300	0.00	3.1	7300	7300	0.00	2.4	7300	7300	0.00	18.4
		$\bar{\mu}_k$	6436	6436	0.00	331.8	6436	6436	0.00	272.2	6436	6436	0.00	327.6	6436	6436	0.00	1277.1
5	20	2	1007	1007	0.00	75.5	1007	1007	0.00	64.3	1007	1007	0.00	148.4	1007	1007	0.00	858.8
		$\bar{\mu}_k$	1007	1007	0.00	217.0	1007	1007	0.00	94.1	1007	1007	0.00	232.4	1007	1007	0.00	1631.0
6	20	3	6931	6931	0.00	116.5	6931	6931	0.00	64.8	6931	6931	0.00	85.2	6931	6931	0.00	200.2
		$\bar{\mu}_k$	6515	6515	0.00	115.5	6515	6515	0.00	104.6	6515	6515	0.00	222.2	6515	6515	0.00	413.9
13	50	3	2435	2365	2.87	lim	2438	2338	3.77	lim	2432	2366	2.71	lim	2447	2363	3.43	lim
		$\bar{\mu}_k$	2432	2365	2.75	lim	2433	2337	3.95	lim	2418	2365	2.19	lim	2431	2359	2.96	lim
14	50	3	10521	9996	4.99	lim	10084	9991	0.92	lim	10049	9999	0.50	lim	10045	9996	0.49	lim
		$\bar{\mu}_k$	9118	9064	0.59	lim	9619	9059	5.82	lim	9125	9063	0.68	lim	9170	9060	1.20	lim
15	50	3	2763	2608	5.61	lim	2741	2606	4.93	lim	2665	2612	1.99	lim	2662	2607	2.07	lim
		$\bar{\mu}_k$	2628	2506	4.64	lim	2633	2497	5.17	lim	2628	2509	4.53	lim	2632	2504	4.86	lim
16	50	4	2796	2693	3.68	lim	2805	2681	4.42	lim	2793	2697	3.44	lim	2799	2690	3.89	lim
		$\bar{\mu}_k$	2794	2645	5.33	lim	2803	2644	5.67	lim	2769	2648	4.37	lim	2817	2643	6.18	lim
17	75	4	1786	1696	5.04	lim	1824	1691	7.29	lim	1802	1698	5.77	lim	1787	1695	5.15	lim
		$\bar{\mu}_k$	1817	1686	7.21	lim	1782	1682	5.61	lim	1776	1686	5.07	lim	1811	1685	6.96	lim
18	75	3	2545	2454	3.58	lim	2646	2455	7.22	lim	2551	2455	3.76	lim	2682	2451	8.61	lim
		$\bar{\mu}_k$	2449	2255	7.92	lim	2402	2249	6.37	lim	2404	2255	6.20	lim	2411	2252	6.59	lim
19	100	4	10639	9720	8.64	lim	10118	9720	3.93	lim	10126	9720	4.01	lim	9811	9720	0.93	lim
		$\bar{\mu}_k$	9371	8588	8.36	lim	9501	8589	9.60	lim	9489	8589	9.48	lim	9351	8588	8.16	lim
20	100	4	4403	4175	5.18	lim	4304	4174	3.02	lim	4402	4173	5.20	lim	4463	4176	6.43	lim
		$\bar{\mu}_k$	4396	3855	12.31	lim	4432	3857	12.97	lim	4366	3855	11.70	lim	4445	3855	13.27	lim

Table 11: Computational results comparing MILP formulations of the HCVRP, strengthened with all violated inequalities from three families at the root node.